Bladder Cancer & Diversion

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 129	Tumours of the Urinary Bladder
Ch 130	Management Strategies for NMIBC
Ch 131	Muscle-Invasive & Metastatic Bladder Cancer
Ch 132	Surgical Management of Bladder Cancer
Ch 133	Intestinal Segments in Urinary Diversion
Ch 134	Continent Urinary Diversion
Ch 135	Minimally Invasive Urinary Diversion
Ch 124	Bladder Surgery for Benign Disease (incl. vesicovaginal fistula)
★	Putting it together

129 Tumours of the Urinary Bladder

1 · Epidemiology (headline)

Why this matters: bladder cancer is common and keeps recurring, so surveillance lasts years.

- There are about 550,000 new cases and 200,000 deaths a year worldwide (2018). It is the **fourth most prevalent** cancer and the **costliest per patient over a lifetime**.
- The median age at diagnosis is **73**, and about 9 in 10 patients are over 55. The **male:female ratio is about 3:1**.
- **Women present later and do worse stage for stage**, because haematuria is often treated as urinary tract infection (UTI).

2 · Risk factors

Why this matters: the exposure predicts the histology. Aromatic amines cause urothelial carcinoma (UC); chronic inflammation causes squamous or glandular tumours.

- P2 CA bladder, etiology, newer investigations and treatment — *risk-factor table (occupation, arsenic, Lynch/MSH2, obesity, diet)*

Not in the paper answer:

- About 20% still smoke at diagnosis, which raises recurrence in non-muscle-invasive (NMIBC) and cancer death in muscle-invasive bladder cancer (MIBC).
- **Radiation**: external beam radiotherapy (EBRT) for cervical cancer raises risk 2–4×, for prostate 1.5–2×. Brachytherapy carries less, intensity-modulated radiotherapy (IMRT) little or none.
- Germline variants occur in 13–24% of UC (MSH2, BRCA1/2). They occur in high-grade, not low-grade, NMIBC, so test selected high-grade patients.
- **Cyclophosphamide** is the only proven chemotherapeutic carcinogen, via **phosphoramide mustard** (†acrolein causes haemorrhagic cystitis, not cancer). Risk is ~4.5×, above 20 g; latency peaks at 10–14 years.
- **Exstrophy**: malignancy 3.3–7.5%, **>90% adenocarcinoma**.
- **Smoking** causes **30–40%** of UC and raises risk **2–3×**. Pipes, cigars and e-cigarettes count too.
- **N-acetyltransferase 2 (NAT2) slow acetylators** carry the highest risk, because acetylation detoxifies aromatic amines. Risk falls to near that of never-smokers **>15 years** after quitting.
- Occupations also include tobacco and petroleum workers. NAT2 and glutathione S-transferase mu 1 (GSTM1) variants matter mainly with exposure.
- ***Schistosoma haematobium*** is the only species linked. Eggs in vesical venules cause T-helper 2 inflammation and fibrosis, which leads to **squamous cell carcinoma (SCC)**; it potentiates smoking. Catheters in spinal cord injury and stones also cause SCC.
- Dietary signals vanish once smoking is controlled, and supplements do not protect.

3 · Pathology

Why this matters: grade and variant histology, not just stage, decide whether the bladder is kept.

Normal urothelium is 4–7 cells thick: basal, intermediate, then **umbrella cells**, which form the urine–blood barrier.

Benign lesions (reactive mimics)

Lesion	Hallmark	Potential / action
Squamous/glandular metaplasia	Trigone; squamous in up to 40% of females	None; no biopsy
Inverted papilloma	Trigone; endophytic nests, no atypia; <1% of tumours	Benign, ~1% recur; resect
Urothelial papilloma	Delicate stalks; fibroblast growth factor receptor 3 (FGFR3) mutant, TP53/RB wild type	None (formerly called low-grade Ta); resect
Nephrogenic adenoma	Tubule-like glands after injury, transplant or intravesical therapy	None; re-resect if it recurs
Leukoplakia	Keratinising squamous metaplasia; floating white flakes	**Not premalignant** (no TP53 evidence), so no follow-up
Cystitis cystica/glandularis	From proliferating von Brunn nests	Glandularis: low adenocarcinoma risk, so resect, treat cause, survey

Flat (precursor) lesions

Loss of polarity means dysplasia; loss of umbrella cells means carcinoma in situ (CIS).

Lesion	Key features	Significance
Hyperplasia	≥10 layers, no atypia	Flat may precede low-grade papillary (FGFR3)
Reactive atypia	Atypia proportionate to inflammation; polarity kept	Coexists with dysplasia/CIS, so investigate
Atypia of unknown significance	Atypia disproportionate to inflammation; CK20, CD44	Limited utility
Dysplasia	Loss of polarity, umbrella cells kept; CK20, p53	14–19% progress; **60%** progress to CIS if there was prior cancer
CIS	Flat, high grade, noncohesive pleomorphic cells; umbrella cells lost	Precursor of invasive high-grade disease

UC grade (2004 World Health Organization, WHO) – noninvasive spectrum

NMIBC makes up about **70%** of UC and includes Ta, CIS and T1.

	PUNLMP*	Low-grade Ta	High-grade papillary	CIS
Morphology	Papillary >7 cells, minimal atypia, solitary, trigone	Branching stalks, some atypia	Fused papillae, many mitoses	Flat, severe atypia
Genetics	TERT promoter, FGFR3	Chromosome 9 loss; FGFR3, phosphoinositide 3-kinase (PI3K), RAS	TP53/RB	RB, TP53, PTEN; **no FGFR3**
Behaviour	Recur ~20% at 5 y; progress <1%	Recur 48–71%; progress 2–12%	>80% invade if untreated	With invasive tumour: 45–65% 5-y mortality

*Papillary urothelial neoplasm of low malignant potential. **Grade, not recurrence, predicts danger.**

Histological variants of UC

About 80% have mixed (mostly squamous) differentiation. **Altered growth patterns behave worse than their stage.**

Variant	Hallmark	Chemo-sensitive?	Management
Micropapillary (0.7–2.2%)	Tight clusters with retraction artefact; **any amount counts**; ERBB2 amplification, luminal	No	**Early radical cystectomy (RC) even for NMIBC** (bacillus Calmette–Guérin (BCG) fails unless fully resected); MIBC immediate RC
Sarcomatoid (0.3–0.6%)	Large infiltrative mass; same clone as carcinosarcoma	Poor	Immediate RC, with or without intraoperative radiation
Plasmacytoid (<1%; includes signet ring)	Eccentric nuclei; **intraperitoneal spread**; CDH1 truncation	Yes, but relapse is common	Neoadjuvant chemotherapy (NAC) whenever possible
Nested (M:F 6:1)	Mimics von Brunn nests and inverted papilloma	No	Often upstaged; stage-matched outcome as UC
Squamous (16–22%, up to 60%)	Basal subtype	NAC active in mixed tumours	Linked to recurrence and progression
Glandular (~10%)	Gland-like spaces	–	Outcome as conventional UC

Non-urothelial malignancy

	Small cell	SCC	Adenocarcinoma	Urachal adenocarcinoma
Setting	Rare; dedifferentiated UC (not lung lineage)	2–5% in West; majority where schistosomiasis endemic	**Secondary is commoner than primary** (colon, prostate, uterus, cervix); exstrophy	~1/3 of bladder adenocarcinomas; arises from the allantoic remnant
Hallmark	Chromogranin, synaptophysin; paraneoplastic (hypercalcaemia, Cushing); TP53, RB1	Locally advanced, fewer metastases	Cribriform/glandular; **colonoscopy** to exclude a primary	Dome/anterior wall; Sheldon staging
Treatment	**Cisplatin–etoposide first**, then RC or radiation	RC; preoperative radiation worsened survival	RC; no level 1 data; 5-fluorouracil (5-FU)–cisplatin by analogy	**En bloc dome with urachal ligament and umbilicus**

KEY POINT

Small cell is systemic disease. Primary cystectomy gives poor disease-free survival (DFS) that adjuvant chemotherapy does not rescue, whereas neoadjuvant chemotherapy markedly improves cancer-specific survival. SCC in spinal cord injury fell after the switch to intermittent self-catheterisation, which shows that inflammation drives it.

Urachal carcinoma (2016 WHO criteria): it sits at the dome or anterior wall with its epicentre in the wall, there is no widespread cystitis glandularis, and a secondary source is excluded. Tumour, node, metastasis (TNM) staging is unsuitable.

Sheldon stage	Extent
I	Urachal mucosa
II	Confined to urachus
IIIA / IIIB / IIIC	Bladder / abdominal wall / peritoneum
IV	Metastatic

4 · Molecular biology

Why this matters: two pathways explain why low-grade tumours recur but rarely kill; subtypes guide drug choice.

	Papillary pathway	Flat pathway
Origin and product	Hyperplasia gives low-grade papillary tumour	Dysplasia gives CIS, then high-grade invasive tumour
Genetics	**FGFR3** (up to 80% of low-grade Ta vs 10–20% of MIBC), chromosome 9 deletion, PIK3CA (25% of NMIBC)	**TP53** (up to 76% of MIBC), RB1, CDKN2A, PTEN loss
Behaviour	Recurs, rarely progresses	Progresses

- HRAS/KRAS mutations are **mutually exclusive with FGFR3** (same pathway) and equally common in NMIBC and MIBC.
- RB1 loss causes E2F3 overexpression. PTEN loss with TP53 alteration leads to metastasis. FGFR3–TACC3 fusions are activated oncoproteins.

The Cancer Genome Atlas (TCGA) subtype	Luminal	Basal/squamous
Markers	FGFR3, GATA3, KRT20, PPARG, FOXA1	CK5, CK6, CK14; low GATA3 and FOXA1
Behaviour	Less aggressive; papillary	Aggressive; squamous; women; T3/T4
Targets	FGFR3 inhibitors, ERBB2	Cisplatin NAC; checkpoint inhibitors; EGFR

- The **consensus classification (2020) has six classes**: luminal papillary (24%; best, median overall survival (OS) 4 y), luminal nonspecified, luminal unstable, stroma-rich, **basal/squamous (35%, the commonest)** and neuroendocrine-like (3%; RB1 loss; worst, ~1 y).
- In TCGA, luminal-infiltrated tumours respond poorly to cisplatin NAC (try immunotherapy); neuronal ones get etoposide–cisplatin; APOBEC-high survives better. MCQ Micropapillary maps to a luminal class.

5 · Clinical presentation and evaluation

Why this matters: about one in five visible haematurias hides a cancer.

- **Painless gross haematuria occurs in 85%**, and microscopic haematuria in nearly all. Irritative symptoms suggest CIS.
- Evaluate every gross haematuria, even if later urinalysis is negative or the patient is anticoagulated.

- Check prostate-specific antigen (PSA): ~10% of recurrent gross haematuria is prostate cancer. Do a bimanual examination at transurethral resection of bladder tumour (TURBT) if invasive *See Ch 132*.
- After a negative work-up, risk is near zero for 6 years; repeat at 3–5 years if haematuria persists.

	Gross haematuria	Asymptomatic microhaematuria
Malignancy	~20%	2.6–5.2%
Cystoscopy	All adults	Age ≥35 years
Upper tract	Computed tomography urogram (CTU)	CTU (American Urological Association, AUA 2012); ultrasound may replace it
Cytology/markers	Reasonable	Avoid routinely

The dipstick detects haem peroxidase: sensitivity 91–100%, specificity ~65%, so **confirm with microscopy**.

Cystoscopy and enhanced imaging

Cystoscopy with biopsy is the gold standard; flexible equals rigid. White light cystoscopy (WLC) misses small papillary tumours and CIS.

	Blue light (HAL–BLC)	Narrow-band imaging (NBI)
Principle	Hexaminolevulinate (HAL) makes porphyrins build up in tumour; they fluoresce red under blue light (360–450 nm)	No dye; 415 and 540 nm light is absorbed by haemoglobin, so vessels stand out dark
Benefit	Finds more Ta and CIS; CIS is seen only on blue light cystoscopy (BLC) in ~¼; office BLC finds ~1/5 of recurrences missed by WLC	Better CIS detection; randomised trial: 12-month recurrence same overall, lower in low-risk
Status	Adjunct; only HAL is Food and Drug Administration (FDA) approved	Adjunct

Urine cytology and biomarkers

- P1 P3 Urine cytology; Asymptomatic haematuria and urinary biomarkers — *good for high grade only; Paris System (HGUC N:C ≥0.7); NMP22, BTA, FISH, ImmunoCyt; none replaces cystoscopy*

> **ALGORITHM**
> 1. Cytology shows high-grade cells but cystoscopy is negative.
> 2. Take random bladder biopsies and prostatic urethral biopsies.
> 3. If these are negative, take selective upper-tract washings for cytology, with or without ureteroscopy.

6 · Staging and risk groups

Why this matters: stage and risk group drive every later treatment decision.

Stage	TNM, American Joint Committee on Cancer (AJCC) 2017
0a / 0is	Ta (noninvasive papillary) / Tis (CIS)
I	T1 – lamina propria (subepithelial connective tissue)
II	T2a inner / T2b outer half of muscularis propria, N0
IIIA	T3a (microscopic perivesical) / T3b (macroscopic) / T4a (prostatic stroma, uterus, vagina) N0; or T1–T4a N1 (single true pelvic node)

Stage	TNM, American Joint Committee on Cancer (AJCC) 2017
IIIB	T1–T4a N2 (≥2 true pelvic nodes) or **N3 (common iliac)**
IVA / IVB	T4b (pelvic/abdominal wall) or M1a (distant nodes) / M1b (organs)

- **The 2017 change**: regional nodes without distant metastasis are now stage III (previously IV). MCQ Common iliac nodes are N3, not M1a.
- **T1a/T1b substaging** (relative to muscularis mucosae) is recommended: ~50% of T1 are upstaged to ≥T2 at cystectomy, and deeper is worse.
- **Prostatic urethra**: mucosal CIS is not adverse in itself but escapes intravesical therapy. Only **prostatic stromal invasion is T4a** (5-year survival historically <25%).

NMIBC risk group	Definition
Low	Primary, solitary, low-grade Ta, <3 cm
Intermediate	Low-grade Ta that is multiple and/or recurrent and/or >3 cm
High	Any high grade, CIS or T1

130 Management Strategies for Non–Muscle-Invasive Bladder Cancer

1 · Definition and natural history

Why this matters: grade decides whether the threat is recurrence or progression.

- **Non-muscle-invasive bladder cancer (NMIBC)** is urothelial carcinoma (UC) that has not invaded the detrusor: Ta, T1 and carcinoma in situ (CIS). It makes up **70–80%** of new bladder cancers: Ta 60–70%, T1 20–30%, CIS ~10% *See Ch 129*.
- Recurrence occurs in **40–60%**; progression in ~6% of low-grade (LG) Ta to 17% of high-grade (HG) T1.
- **Grade, not stage, is the strongest predictor of progression**, and CIS comes second. HG Ta and HG T1 progress similarly. Primary CIS progresses in **>50%**, and untreated CIS reaches muscle in 40–83%.
- **Understaging**: cystectomy for presumed NMIBC shows worse disease in **one-third to one-half**; clinical T1 is unreliable.

> **KEY POINT**
> NMIBC is two diseases. LG disease is a **recurrence** problem; HG disease is a **progression** problem. Chemotherapy treats recurrence, bacille Calmette-Guérin (BCG) treats progression, and cystectomy is for progression that cannot be controlled.

2 · Risk stratification

*Why this matters: the risk group sets the regimen and surveillance. Risk is **dynamic**, so reassess it at every recurrence.*

American Urological Association/Society of Urologic Oncology (AUA/SUO) 2016 (updated)

Risk	Criteria	Intervention
Low	LG solitary Ta **≤3 cm**; papillary urothelial neoplasm of low malignant potential (PUNLMP)	Single postoperative chemotherapy instillation
Intermediate	LG Ta recurring within 1 yr; LG Ta solitary >3 cm or multifocal; **LG T1**; **HG Ta ≤3 cm**	6-week induction of chemotherapy or BCG; maintenance to 1 yr
High	HG T1; recurrent HG Ta; HG Ta >3 cm or multifocal; **any CIS**; any BCG failure in HG; **any variant histology**; **any lymphovascular invasion (LVI)**; any HG prostatic urethral involvement	Induction BCG and maintenance to **3 yr**. If BCG-unresponsive: radical cystectomy (RC), pembrolizumab, nadofaragene or gemcitabine–docetaxel

EORTC and CUETO scoring models

European Organisation for Research and Treatment of Cancer (EORTC); Club Urológico Español de Tratamiento Oncológico (CUETO).

Factor	EORTC	CUETO
Number, prior recurrence rate, T stage, concurrent CIS, grade (World Health Organization (WHO) 1973)	✓	✓
Tumour diameter	✓	—
Age, sex	—	✓

- EORTC gives 1- and 5-year risks: **recurrence** from prior rate, number and size; **progression** from T stage, CIS and grade.
- EORTC **overestimates** risk, because its patients had little maintenance BCG, perioperative mitomycin C (MMC) or re-TURBT. CUETO, derived in BCG-treated patients, fits better. MCQ Both use **WHO 1973**.

3 · Endoscopic management

Why this matters: complete resection with muscle in the specimen underpins correct staging.

TURBT: principles

- Transurethral resection of bladder tumour (TURBT) is both diagnostic and therapeutic (technique *See Ch 132*). Its quality levers are **complete resection** and **detrusor in the specimen**; missing muscle means higher mortality.
- Image the upper tracts at diagnosis by computed tomography urography or retrograde pyelography. Upper tract tumours occur in **<5%**; risks are trigonal tumour, CIS and high risk.
- **Diverticular tumour**: there is no detrusor, so invasion beyond the lamina propria is **T3a**. LG: cold-cup and fulguration. HG: strongly consider partial cystectomy or RC.
- **En bloc resection** gives fewer perforations, more detrusor and better staging, but the **same recurrence**.

Repeat (second-look) TURBT

- P4 Re-look (second-look) TUR bladder tumour — *indications (all T1, incomplete, no muscle; consider HG Ta), within 6 weeks; residual ≥40%, HG T1 upstaged 25–30% (~50% without muscle); residual T1 means 80% progression*

Recommended (within 6 weeks, include muscularis propria)	Consider
Incomplete resection; all T1 (especially without muscle); no muscle in specimen	High-risk HG Ta (multifocal and/or large)

Biopsies and office management

- Cold-cup biopsies avoid cautery artefact. Random biopsies are **not** needed in low risk; consider them with positive cytology or high risk. Biopsy the prostatic urethra if there is CIS or a bladder neck/trigone tumour.
- Small (**≤1 cm**) LG recurrences can be fulgurated or lasered in the office. No tissue is obtained, so this needs **prior LG diagnosis and negative cytology**. **Active surveillance** of small LG tumours is safe in selected patients.

Enhanced cystoscopy

	Blue-light (hexaminolevulinate, HAL)	Narrow-band imaging (NBI)
Principle	Porphyrins fluoresce red under blue light; needs intravesical instillation	415 nm (superficial) and 540 nm (deeper) light is absorbed by haemoglobin; no agent
Gain	~**one-third more CIS**; smaller papillary tumours	More papillary tumours and CIS
Weakness	Cost; a pragmatic randomised trial showed no 3-yr recurrence benefit	False positives ~60%

- HAL replaced 5-aminolevulinic acid (no systemic photosensitisation). Progression benefit is unproven; AUA: use if available.

4 · Single post-operative instillation

Why this matters: one early dose kills floating cells before they implant, which cuts recurrence.

- P3 Immediate post TURBT intravesical Therapy for TCC — *anti-implantation; within 6 h (useless at 24 h); MMC, epirubicin, gemcitabine 2 g (SWOG S0337); best in solitary LG; withhold if perforation; never BCG*
- Give MMC within **6 h**; the evidence is level 1. The gain is limited in recurrent or multiple tumours. Thiotepa is ineffective.
- After perforation it risks chemical cystitis, desquamation, contracture and rare deaths.

5 · Intravesical chemotherapy

Why this matters: chemotherapy is the main tool against intermediate-risk recurrence.

- P4 Intravesical therapy for urothelial (transitional cell) carcinoma of bladder, including combination intravesical therapy — *MMC optimisation (Au 2001); Gem/Doce regimen; device-assisted; BCG mechanism, schedule, toxicity*

Not in the paper answer:

- Valrubicin achieves CR in only ~1/5 of BCG-refractory CIS; MMC contracts 5% of bladders.
- After BCG, only ~30% with CIS are free of recurrence or progression at 10 yr.
- A quinolone for 24–48 h after each BCG dose reduces cystitis and delays without loss of efficacy. TICE is the only FDA-approved strain.
- Chemotherapy reduces **recurrence**, with **no proven effect on progression**. In intermediate risk, MMC or gemcitabine equals BCG, especially in LG multifocal recurrent disease. Give 6-week induction, then monthly for 1 yr.

Agent	Class	Dose	Signature toxicity
Mitomycin C	Alkylating	40 mg/20 mL weekly ×6	Rash, severe skin reaction, contracted bladder 5%
Gemcitabine	Nucleoside analogue	2 g/100 mL	Mild; minimal absorption
Doxorubicin / epirubicin	Anthracycline	50 mg/50 mL	Chemical cystitis, contracted bladder
Valrubicin	Anthracycline	800 mg/75 mL	Food and Drug Administration (FDA) approved for BCG-refractory CIS; complete response (CR) ~1/5
Thiotepa	Alkylating	60 mg/30 mL	Myelosuppression
Docetaxel	Taxane	75 mg/100 mL	Myelosuppression

- Electromotive drug administration (EMDA) and chemohyperthermia (Synergo, hyperthermic intravesical chemotherapy (HIVEC)) beat passive MMC.
- **Gemcitabine–docetaxel (Gem/Doce)** includes maintenance in responders (regimen in P4-068). It is increasingly used in BCG-naive HG; a phase III trial against BCG is under way.

6 · Bacille Calmette-Guérin (BCG)

Why this matters: BCG prevents progression, but only with maintenance.

Mechanism

BCG binds fibronectin, which releases cytokines and draws in immune cells, giving **T-helper 1 (Th1)** cytotoxicity (see P4-068). Interleukin-10 drives a suppressive Th2 response. No strain is superior; age and immunosuppression blunt response.

Administration

ALGORITHM

1. Wait **2–4 weeks** after TURBT (re-epithelialisation).
2. Check urinalysis before each dose (exclude infection, gross bleeding).
3. Instil 1 vial in 50 mL saline by catheter.
4. Retain 1–2 h; restrict fluids, diuretics and caffeine beforehand.
5. If catheterisation is traumatic, delay several days.

Contraindications

Absolute	Relative
Immediately after TUR; gross haematuria; traumatic catheterisation; previous BCG sepsis; immunosuppression*; total incontinence	Urinary tract infection (UTI); liver disease (isoniazid cannot then be given for sepsis); previous tuberculosis; anti-tumour necrosis factor (TNF) drugs; poor performance status; advanced age

- Most absolute contraindications mean a **breached urothelium, which allows intravasation**. *Safe in transplant and steroids, but the response is weaker.

Efficacy

- BCG is the **only intravesical agent that reduces progression**, and only in trials with **maintenance**; it beats MMC only with maintenance.
- In CIS, the initial CR is up to **84%**, and about half are durable.
- A second induction for persistent CIS gives a further 30–50% CR. Beyond 2 courses, progression risk is 20–50% (~7% per course). **The response at 6 months predicts prognosis.**

Schedule

ALGORITHM

1. **SWOG (Southwest Oncology Group; Lamm 2000)**: give induction weekly ×6.
2. Then give 3 weekly instillations at **3 and 6 months, then every 6 months to 3 yr**.
3. If high risk, give full dose for 3 yr. If intermediate risk, give 1 yr (EORTC 30962).

Toxicity management

Grade	Features	Management
1	Cystitis, mild haematuria, fever <38.5°C, lasting <48 h	Symptomatic; consider culture
2	Severe symptoms or >48 h	Culture, chest X-ray, liver tests; isoniazid, rifampicin and pyridoxine; resume at ½–⅓ dose
3	Haemodynamic change, persistent fever; allergic (arthralgia, rash); organ involvement (liver, lung, kidney)	**Stop BCG**; isoniazid, rifampicin and ethambutol; prednisone only **with** antimycobacterial cover; infectious disease input

- MCQ BCG is resistant to **pyrazinamide**. Avoid cycloserine (psychiatric effects). Asymptomatic granulomatous prostatitis needs no treatment.

BCG shortage

ALGORITHM

1. Prioritise high-risk patients.
2. Reduce the dose, especially in maintenance, and split vials.
3. Use chemotherapy: single agent for intermediate, Gem/Doce for high risk. Early RC for the highest risk.

7 · Recurrence after BCG

Why this matters: the failure category decides between more BCG and cystectomy.

- P2 BCG failure — *definitions table: adequate BCG (≥5/6 induction + ≥2/3 maintenance; T1 induction alone), unresponsive (HG within 6 months Ta/T1, 12 months CIS), resistant (reinduce, CR 30–50%), intolerant, delayed relapse*

Not in the paper answer:

- After failure, check the prostatic urethra and upper tracts: urethral or upper tract recurrence develops in ~half on long follow-up.
- Radiotherapy is palliative and has no effect on CIS.

KEY POINT

Trap: HG T1 at 3 months after induction alone counts as unresponsive. CIS or HG Ta at 3 months is "resistant", so give more BCG.

Bladder-sparing options (unfit for or declining RC)

Agent	Mechanism / schedule	CIS CR at 3 mo, then 12 mo	Cystectomy-free
Pembrolizumab (KEYNOTE-057)*	Anti-programmed death-1 (PD-1); 200 mg intravenously every 3 weeks up to 2 yr	41%, then 19%	63%; grade 3–4 toxicity 13% (immune)
Nadofaragene firadenovec*	Non-replicating adenovirus carrying the interferon-α2b gene, with Syn3; intravesical every 3 months	53%, then 24%	71%; grade 3–4 4%
N-803 + BCG	Interleukin-15 superagonist that activates CD8+ T and natural killer cells	71% (any time), then 45%	~85%
Gem/Doce	Weekly ×6, monthly ×2 yr	—, then 60% (43% at 2 yr)	84%; HG recurrence-free survival (RFS) 65%/52% at 1/2 yr; no level 1 evidence

*FDA-approved. CR roughly halves by 12 months, and none matches cystectomy.

8 · Early radical cystectomy in high-risk NMIBC

Why this matters: delay with failing intravesical therapy costs lives.

- T1 failing BCG at 3 months progresses in **82%**, and CIS in ~50%. With prolonged intravesical therapy, only ~¼ are alive with their bladder at 15 yr.
- Immediate RC for clinical T1 gives 10-yr disease-free survival of **92%** (64% if understaged). Even a 12-week delay worsens survival.

- **Patients who progress to muscle invasion under surveillance do worse than those with de novo muscle-invasive disease.**

ALGORITHM

1. **Strongly consider RC** if: BCG-unresponsive HG; persistent HG T1 on re-resection; T1 with CIS, LVI or variant histology; deep lamina propria invasion; diffuse CIS.
2. Also if: diverticular HG tumour; distal ureteric or prostatic ductal involvement; incompletely resectable tumour; patient preference.

9 · Surveillance

Why this matters: NMIBC recurs for years.

Cystoscopy with cytology is the gold standard. **Tumour at the 3-month cystoscopy** and number at first resection best predict recurrence.

Risk	Cystoscopy	Cytology	Upper tract imaging
Low	At 3 months, then annually from 12 months; consider stopping at **≥5 yr**	Consider	Only if haematuria
Intermediate	3–6-monthly for 1–2 yr, then 6–12-monthly for 2 yr, then annually	Consider	Consider every 1–2 yr
High	**3–4-monthly for 2 yr**, then 6-monthly for 2 yr, then annually **for life**	Same schedule	Annually for 2 yr, then lengthen

- Restart the clock at each recurrence, using the risk of the **latest resected** tumour.
- Persistent positive cytology with negative cystoscopy and imaging needs biopsies, including prostatic urethra, and upper-tract work-up.
- **UroVysion fluorescence in situ hybridisation (FISH)** is the most specific marker; an "anticipatory positive" can precede tumour. AUA: markers do not replace cytology; use them for BCG response or atypical cytology.
- **Extravesical recurrence**: upper tract UC occurs in <1% at low risk and ~10% at high risk (13–18% after BCG), and the risk persists ≥15 yr. Prostatic urethra is involved in 20–40% of high risk at 10 yr; HG ductal disease needs cystoprostatectomy with or without urethrectomy.

10 · Prevention

Why this matters: smoking advice belongs at every visit.

- **Advise stopping smoking** (effect on recurrence uncertain), more fluids and a low-fat diet.

11 · Landmark trials

Why this matters: these trials set today's standards.

Trial	Design	Result	Take-home
SWOG S0337	Immediate gemcitabine 2 g vs saline, suspected LG	4-yr recurrence 35% vs 47% (LG 34% vs 54%); no extra toxicity	Gemcitabine is the standard single instillation
SWOG maintenance (Lamm 2000)	Induction with or without maintenance BCG, recurrent Ta/T1/CIS	Median RFS 76.8 vs 35.7 months; only 16% completed	Maintenance is essential

Trial	Design	Result	Take-home
EORTC 30962	Full vs ⅓ dose; 1 vs 3 yr	Toxicity stops ~7–8%	Intermediate: full dose 1 yr; high: full dose 3 yr
NIMBUS	Reduced-frequency (9 doses) vs standard, HG	Recurrence doubled; stopped early	Do not reduce frequency
KEYNOTE-057	Single-arm pembrolizumab, BCG-unresponsive CIS	CR 41% (3 mo), 19% (12 mo)	First systemic therapy approved for NMIBC

131 Management of Muscle-Invasive and Metastatic Bladder Cancer

1 · Natural history and histology

Why this matters: MIBC kills through micrometastases, so cure needs systemic as well as local treatment.

- **20–30%** of new bladder cancers are muscle-invasive (MIBC), and **~12%** are node-positive or metastatic.
- **Primary MIBC** is invasive at diagnosis; **secondary** (**~20%**) has progressed from non-muscle-invasive disease (NMIBC). It does worse: more understaging and less response to neoadjuvant chemotherapy (NAC).
- Most deaths after local therapy occur within **2 years** and come from micrometastases present at diagnosis.
- Urothelial carcinoma (UC) is 90%. **Radical cystectomy (RC) is standard whatever the histology**; subtype changes only NAC and radiation use. Variants are upstaged more often.

Variant	Key points	Treatment
Squamous (5%)	Schistosomiasis, chronic cystitis	Poor chemo- and radiosensitivity, so up-front RC with lymphadenectomy
Adenocarcinoma (2%)	Commonest in exstrophy; urachal; exclude colorectal or breast metastasis	RC (urachal: partial cystectomy)
Small cell	High stage; ectopic adrenocorticotropic hormone (ACTH), hypercalcaemia	**Chemosensitive**, so NAC with etoposide and cisplatin, then RC
Micropapillary	Frequent lymphovascular invasion (LVI)	If predominant, up-front RC
Mixed UC + variant	Quantify the variant	UC-predominant: NAC; variant-predominant: RC

2 · Clinical staging

Why this matters: clinical staging often underestimates cT2 disease.

Stage in order: transurethral resection (TUR) depth, examination under anaesthesia (EUA), computed tomography (CT) with excretory phase, then chest imaging. In cT2, pathology upstages **40–54%** and downstages ~17%. LVI, variant histology, female sex and secondary MIBC predict upstaging.

TURBT

- Resect all visible tumour if safe, with a separate deep specimen. This may aid NAC response and is needed for preservation; pT0 at RC improves survival even without NAC.
- If high-grade T1 or no muscle, repeat TUR within **6 weeks** (>40% are upstaged without muscle).
- **Prostatic urethral biopsy** (trigone or bladder-neck tumour, carcinoma in situ (CIS), multifocal): take a full-thickness loop from mid-prostate to verumontanum at **5 and 7 o'clock**, where ducts are densest. Stromal involvement affects NAC, urethrectomy and diversion choice.

- **Frozen urethral margin at cystoprostatectomy** is mandatory if the TUR biopsy was positive (neobladder only if negative). Do it anyway if a neobladder is planned (higher negative predictive value).

Bimanual EUA

Examine before and after resection, bladder empty. EUA overstages and, more often, understages.

cT	Bimanual finding
T2a	Nonpalpable
T2b	Induration, no three-dimensional mass
T3a	Mobile three-dimensional mass
T4a	Invades prostate, vagina, rectum
T4b	Fixed to pelvic sidewall

Imaging and bloods

- **Image before TUR**, as post-resection inflammation mimics T3. CT nodal sensitivity is only **30–50%**; hydronephrosis suggests extravesical disease.
- Magnetic resonance imaging (MRI) suits poor renal function or contrast allergy and stages T best. The **Vesical Imaging-Reporting and Data System (VI-RADS)** grades the muscularis line: intact in 1–3, interrupted in 4, into fat in 5. A score **≥3** is suspicious, so "equivocal" counts as positive.
- Fluorodeoxyglucose positron emission tomography (FDG PET)/CT is not routine; the American Urological Association (AUA) uses it for equivocal imaging when biopsy is impossible. It is useless locally (urinary excretion).
- Chest CT for all; bone or brain imaging only with symptoms or raised alkaline phosphatase. Renal function decides NAC and orthotopic diversion.

3 · Pathological staging (AJCC 8th) and prognostic markers

Why this matters: pathological stage decides adjuvant therapy.

Category (American Joint Committee on Cancer)	Definition
T2a / T2b	Inner / outer half of muscularis propria
T3a / T3b	Perivesical fat microscopic / macroscopic
T4a / T4b	Prostatic stroma, seminal vesicles, uterus, vagina / pelvic or abdominal wall
N1 / N2 / N3	Single true-pelvis node (perivesical included) / multiple / **common iliac**
M1a / M1b	Nodes beyond common iliacs / non-nodal
Stage II / IIIA / IIIB	T2N0 / T3–T4a N0 or T1–T4a N1 / N2–N3
Stage IVA / IVB	T4b any N, or M1a / M1b

- Traps: common iliac is **N3, not M1**; T4b N0 is stage IVA. pT4a needs **stromal** invasion (urethral or ductal CIS does not upstage). Organ-confined is ≤pT2bN0.

- **Subtypes**: luminal (fibroblast growth factor receptor 3 (FGFR3) mutations) survives best; **basal gains most from NAC**; p53-like is chemoresistant. They do not yet guide treatment.
- **Circulating tumour DNA (ctDNA)**: >⅓ stay positive after RC and do badly; in IMvigor010, adjuvant atezolizumab helped only them.

4 · Radical cystectomy: principles and extent

Why this matters: RC is the standard cure; organ sparing must not compromise it.

- **RC with bilateral pelvic lymph node dissection (PLND) is standard for resectable MIBC**, ideally after cisplatin NAC. Delay > **12 weeks** means more extravesical and nodal disease and worse survival.
- **Men**: bladder, perivesical tissue, prostate, seminal vesicles. **Women** (anterior exenteration): bladder, urethra, uterus, cervix, ovaries, tubes, anterior vaginal wall. Steps *See Ch 132*; diversion *See Ch 133–135*.
- **Male organ-sparing** needs no bladder-neck disease and negative prostatic biopsies; incidental prostate cancer is found in ¼–½, so select carefully.
- **Female organ-sparing** needs no base or trigone tumour. Keeping uterus and anterior vagina supports the neobladder and reduces prolapse. Keep ovaries premenopause unless breast/ovarian cancer risk, and the urethra unless the bladder neck is involved.

> **KEY POINT**
> Spare an organ only where it is oncologically silent. The gatekeepers are, in men, the prostatic urethra and bladder neck; in women, the bladder neck, base and anterior vaginal wall.

5 · Pelvic lymph node dissection

Why this matters: nodal status is the strongest predictor of survival.

- P4 Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer? — *templates; German trial and S1011 negative, so standard PLND; extranodal extension*

Not in the paper answer:

- About **25%** are node-positive at RC. Recurrence is **70–80%** if node-positive vs ~30% if node-negative.
- The obturator and internal iliac nodes are the sentinel stations. Skip metastases are rare, but common iliac nodes are involved in 7–13%.
- About 25 nodes give ~75% detection. A therapeutic node threshold is unproven; 10 nodes is the usual pelvic-recurrence cut-off.
- Node positivity rises with T stage: ≤pT1 1–7%, pT4 up to 80%.
- Borders: ureter medially, genitofemoral nerve laterally, node of Cloquet caudally. Cranially, standard PLND reaches the common iliac bifurcation or ureteric crossing; extended reaches the aortic bifurcation (up to the inferior mesenteric artery is controversial).

6 · Intraoperative decision making

Why this matters: theatre decisions on nodes, margins and urethra change cure and diversion.

- If **clinically node-positive**, give cisplatin chemotherapy first, then **consolidative RC** in responders; 14–25% reach pathological complete response and survive much better.

- **Unexpected nodes**: send frozen section; proceed only if completely resectable within the true pelvis.
- **Abort RC** if nodes are unresectable, periureteral disease is extensive, the bladder is fixed to the sidewall or the rectosigmoid is invaded. Most then die within 2 years.
- **Ureteric margin** is involved in 6–8%; if positive, upper-tract recurrence (UTR, 4–10%, mostly at 2–4 years) is likelier. Frozen section is not mandated; send one for gross tumour. For CIS (skip lesions), resect maximally but keep diversion length. "Converted" margins still carry higher UTR, as do prior bacille Calmette-Guérin (BCG), CIS and hydronephrosis.
- **Male urethral recurrence** is 1–8%, mostly within 2 years, and highest with **prostatic stromal invasion** (~30%); consider urethrectomy then and for diffuse prostatic urethral CIS. A positive pre-op prostatic biopsy does **not** bar a neobladder; the **frozen urethral margin decides** (specificity ~100%). Low-grade papillary urethral tumour: resect before RC, keep the urethra.
- **Female urethra** is involved in 2–12% (bladder neck, base or anterior vaginal wall tumour). Urethrectomy is mandatory only for T4 involving urethra or vagina. Send proximal urethral frozen section in every neobladder candidate.

7 · Outcomes after RC

Why this matters: stage and nodes predict who relapses.

Stage	5-year disease-specific survival
≤pT1 / pT2 / pT3 / pT4	~90% / 70–81% / 40–52% / 16–44%
N0 / N+	62–80% / 21–35%

- **~50%** ultimately die of bladder cancer, mostly from metastases within 2 years.
- **Pelvic recurrence**: low risk (≤pT2) 5–8%; intermediate (pT3–4, negative margins, ≥10 nodes) 15–21%; high (pT3–4 with positive margins or <10 nodes) **39–46%**.

8 · Neoadjuvant therapy

Why this matters: cisplatin NAC gives a level 1 survival gain, but half cannot have it.

- P2 Muscle-invasive bladder cancer (TCC): management, neoadjuvant and adjuvant chemotherapy, neobladder types — *cisplatin NAC (ddMVAC 3–6/GC 4), rationale, pT0 ~90% RFS; adjuvant chemo (ABC 2022 +6%), nivolumab, Cairo radiotherapy*

Not in the paper answer:

- About half of patients are cisplatin-ineligible; platinum-ineligibility criteria are stricter (table).
- Neoadjuvant immunotherapy (IO), phase II: **ABACUS** (atezolizumab) pathological complete response 31%; **PURE-01** (pembrolizumab) pT0 42%.

Criterion	Cisplatin-ineligible (any one; ~50%)	Platinum-ineligible
Renal	Glomerular filtration rate (GFR) **<60 mL/min**	Creatinine clearance <30 mL/min
Eastern Cooperative Oncology Group (ECOG) status	≥2	≥2 (text) / ≥3 (table)
Hearing loss / neuropathy	Grade ≥2 / ≥2	— / grade ≥3
Heart failure	New York Heart Association (NYHA) III–IV	NYHA III–IV; multiple high-grade comorbidities

- The National Comprehensive Cancer Network (NCCN) prefers dose-dense methotrexate, vinblastine, doxorubicin and cisplatin (ddMVAC). Cisplatin-ineligible patients go straight to local therapy.
- **Neoadjuvant IO**: tumour PD-L1 binds programmed death-1 (PD-1) on T cells and switches off killing; blockade restores it. Anti-PD-L1: atezolizumab, durvalumab, avelumab; anti-PD-1: pembrolizumab, nivolumab. The source did not yet recommend IO (NIAGARA, KEYNOTE-866, ENERGIZE pending). †NIAGARA has since shown that perioperative durvalumab with gemcitabine–cisplatin (GC) improves event-free and overall survival (OS).
- Neoadjuvant radiation downstages but gives no survival gain.

9 · Adjuvant therapy

Why this matters: high-risk pathology after RC needs further treatment.

- The target is **pT3–4 or N+**. Post-RC morbidity blocks delivery in ~¼, so neoadjuvant is preferred (compliance, level 1).
- **Chemotherapy** is not standard, but discuss ddMVAC or GC in **cisplatin-naive** pT3–4/N+ patients.
- **Nivolumab for 12 months**: ≥pT3/N+ after NAC, or if cisplatin-ineligible or declining. IMvigor010 (atezolizumab) was negative (included pT2).
- **Radiation**: pT3–4 local failure is 20–40%; median survival after pelvic failure ~9 months. May consider for pT3/N+.

10 · Bladder preservation

Why this matters: selected patients keep their bladder with survival similar to cystectomy.

Trimodality therapy (TMT)

- P4 Discuss bladder preservation surgery for TCC of urinary bladder — *selection (cT2, no hydronephrosis/CIS; 6–19%), sensitisers, schedules, outcomes, salvage RC 10–30%, follow-up; partial cystectomy*

Not in the paper answer:

- Concurrent chemotherapy beats radiation alone (two randomised trials).
- If unfit for RC, criteria are looser; extensive CIS and a poor bladder are relative contraindications.
- TMT needs a multidisciplinary team, a compliant patient and salvage-RC counselling; give fit patients a balanced TMT vs RC discussion.
- **Dose**: a continuous course to **60–66 Gy** with restaging at 3–4 months is commonest. Split course is historical. **55 Gy in 20 fractions** is non-inferior.

> **KEY POINT**
> TMT contains cystectomy rather than replacing it. Salvage RC is the built-in safety net, so selection and lifelong cystoscopy decide success. Complete TUR cuts the need for cystectomy (~11% vs 42%).

Single modalities (none is primary therapy)

Modality	Use	Why it fails
Chemotherapy alone	Complete responders declining RC	Early intravesical recurrence (~half by 5 months); in RETAIN, half recurred despite DNA-repair mutations

Modality	Use	Why it fails
Radiation alone (64–66 Gy/32 or 55 Gy/20 fractions)	Unfit for surgery and chemoradiation; palliation	Inferior to chemoradiation; local control 27% in T3
Radical TUR	Superficial muscle invasion only	Understaging; ~25% have occult nodes
Partial cystectomy with PLND	<3 cm solitary, no CIS, margin away from orifices; **urachal adenocarcinoma (treatment of choice)**	Late recurrence, so lifelong follow-up. Use frozen sections; convert to RC if the margin is positive; consider perioperative cisplatin

11 · Metastatic urothelial carcinoma

Why this matters: first-line choice hinges on platinum eligibility.

- Bajorin adverse factors: Karnofsky performance status <80% and visceral metastasis (median survival 33, 13 and 9 months with 0, 1 and 2).
- **Cisplatin-eligible**: GC or ddMVAC; most respond, but most progress. **GC equals MVAC** and is less toxic (von der Maase). Dose-dense beat standard MVAC (EORTC 30924).
- **Cisplatin-ineligible, platinum-eligible**: gemcitabine–carboplatin (EORTC 30986) or enfortumab vedotin (EV) with pembrolizumab (NCCN preferred). †EV-302 later showed EV with pembrolizumab ~doubled OS vs platinum in all comers, making it the preferred first line.
- **Platinum-ineligible**: pembrolizumab (KEYNOTE-052) or atezolizumab if PD-L1-positive (IMvigor210). Trap: **first-line IO alone is only for platinum-ineligible patients.**
- **Maintenance avelumab** if no progression after platinum (JAVELIN Bladder 100). IO with chemotherapy (IMvigor130, KEYNOTE-361, DANUBE) is not recommended.
- **Second line**: **pembrolizumab** (KEYNOTE-045, regardless of PD-L1); alternatives are nivolumab and avelumab. Atezolizumab failed (IMvigor211). Platinum rechallenge if the last dose was ≥12 months earlier. Vinflunine: Europe only.
- **Antibody–drug conjugates (ADCs)**: **EV** (targets Nectin-4, payload monomethyl auristatin E; neuropathy, rash) beat chemotherapy after platinum and IO (EV-301). **Sacituzumab govitecan** (targets Trop-2, payload SN-38; neutropenia) is third line.
- **Erdafitinib** (oral FGFR inhibitor) treats FGFR3/FGFR2 alterations after platinum: response 40%; **hyperphosphataemia**, stomatitis.
- Bevacizumab, sunitinib and trastuzumab failed.
- **IO toxicity**: rash, colitis, hepatitis, pneumonitis, endocrinopathies (often **irreversible**), myocarditis.

ALGORITHM

1. If platinum-eligible, give GC or ddMVAC (or gemcitabine–carboplatin, or EV with pembrolizumab).
2. If there is response or stable disease, give avelumab maintenance.
3. On progression, give pembrolizumab, or erdafitinib if FGFR-altered.
4. Then give EV, then sacituzumab govitecan.
5. If platinum-ineligible, give pembrolizumab, then EV.

12 · Follow-up

Why this matters: most recurrences come early, but upper-tract ones late.

- **After RC**: CT with excretory phase and chest CT 6-monthly for 2–3 years, then yearly to 5 (European Association of Urology (EAU), AUA; NCCN 3–6-monthly for 2 years). Renal function each visit, vitamin B12 yearly from year 2, cytology 6–12-monthly.
- Urethral wash (controversial) is for CIS, prostatic urethral tumour or bloody discharge.
- Upper-tract recurrence (4–10%) is the commonest **late** one. Pelvic and distant recurrences come mostly within 2 years; pelvic recurrence gets chemoradiation, with poor results.
- Diversion checks: acidosis, renal function, stoma, hernia, stones *See Ch 133–135*.
- **After bladder preservation**: cystoscopy 3–4-monthly in years 1–2, 6-monthly in years 3–4, then yearly to **10 years**. Get CT urogram and chest CT 3–6-monthly for 2 years, then yearly to 5.

13 · Landmark trials

Why this matters: these trials define the pathway.

Trial	Design	Result	Take-home
SWOG 8710	MVAC ×3, then RC, vs RC	Median OS 77 vs 46 months; pT0 38% vs 15%	pT0 is a surrogate; OS P = .06
BA06 30894 (EORTC/MRC)	Cisplatin, methotrexate, vinblastine (CMV) ×3, then RC or radiotherapy	10-year OS 36% vs 30%	Largest positive NAC trial
ABC meta-analysis	11 randomised NAC trials	OS hazard ratio (HR) 0.86; +5% at 5 years	Combination platinum only
VESPER	ddMVAC ×6 vs GC ×4	Organ-confined 77% vs 63%; 3-year progression-free survival 66% vs 56%	ddMVAC better but more toxic
SWOG S1011	Standard vs extended PLND	No difference in disease-free survival (DFS) or OS	Standard PLND suffices
CheckMate 274	Adjuvant nivolumab vs placebo, pT3–4/N+	DFS 20.8 vs 10.8 months	Adjuvant standard
BC2001	Radiation with or without mitomycin C/5-fluorouracil	Locoregional DFS 67% vs 48%	Sensitise radiation
von der Maase	GC vs MVAC, metastatic	OS 14.0 vs 15.2 months (not significant)	Equal efficacy, GC less toxic
EORTC 30986	Gemcitabine–carboplatin vs MCAVI, cisplatin-ineligible	Equal OS, less toxicity	Gem–carbo standard
JAVELIN Bladder 100	Avelumab maintenance vs none after platinum	OS 21.4 vs 14.3 months (HR 0.69)	Maintenance standard
KEYNOTE-045	Pembrolizumab vs chemotherapy, second line	OS 10.3 vs 7.4 months	Second-line standard
EV-301	EV vs chemotherapy after platinum and IO	OS 12.9 vs 9.0 months	ADC third line

132 Surgical Management of Bladder Cancer: Transurethral, Open and Robotic

1 · Framing

Why this matters: the first resection sets the stage.

- Transurethral resection (TUR) stages every patient and cures low-grade non-muscle-invasive bladder cancer (NMIBC).
- Muscle-invasive bladder cancer (MIBC) needs radical cystectomy (RC) and pelvic lymph node dissection (PLND), after cisplatin neoadjuvant chemotherapy (NAC) if eligible *see Ch 131*.

2 · Transurethral resection of bladder tumour (TURBT)

Why this matters: complete resection with muscle in the specimen makes staging correct.

Pre-operative evaluation

- **Continue aspirin**. Hold warfarin, heparins, direct oral anticoagulants (DOACs) and P2Y12 inhibitors by half-life; agree bridging beforehand.
- Do a bimanual examination, and a digital rectal examination (DRE) if prostate is suspected. **Treat active infection before resection**.
- Image the upper tracts by CT urogram, MRI, or ultrasound with retrograde study. Iodinated contrast needs a glomerular filtration rate (GFR) **≥60** mL/min; gadolinium needs GFR **>30**.
- Orifice tumour: have ureteroscope, stent and fluoroscopy ready.

Set-up and inspection

ALGORITHM

1. Lithotomy; antibiotic within 1 h (avoid fluoroquinolones).
2. 30° lens: urethra, orifices. 70° lens: bladder neck, dome, anterior wall.
3. If inspection is incomplete, use the 120° lens.
4. Map tumours; add blue-light or narrow-band imaging if available (more tumours found, less recurrence, detects carcinoma in situ (CIS)).

	Monopolar	Bipolar
Irrigant	Sterile water or 1.5% glycine	0.9% saline
Electrolyte disturbance	Higher risk	Lower risk
Obturator reflex	Present	Reduced

Resection principles

- Remove all visible tumour with **detrusor in the specimen**, using a loop on cutting current, with a **~2 cm** margin of normal tissue where possible.
- Then examine under anaesthesia (EUA), bladder empty: a **mass palpable after resection is cT3b**.
- † **En bloc resection** takes the tumour in one piece, so orientation and detrusor are kept and T1 staging improves.
- Maximal resection is therapeutic in NMIBC; before RC for MIBC it gives no survival gain (residual tumour is only prognostic), as clinical staging is unreliable.

Difficult locations

Site	Hazard	Countermeasure
Lateral wall	**Obturator jerk**: nerve stimulation makes the leg adduct and the loop perforates	Low distension; bipolar; muscle relaxant (general anaesthesia)
Ureteric orifice	Coagulation causes stricture and hydronephrosis	**Pure cutting current**; routine stent not needed; short-term stent after large resection or heavy cautery is reasonable
Dome	Intraperitoneal perforation; heat injury to bowel	Low distension; suprapubic pressure; straight loop; minimal cautery

Bladder perforation

	Extraperitoneal	Intraperitoneal
Site / frequency	Away from dome; commoner	Dome, posterior wall; rare, large ones morbid
Clues	—	Bladder will not distend; defect seen; abdomen distends
Confirm	Cystogram at same sitting	
Management	**Foley drainage and observation**	**Laparotomy**, inspect bowel, repair; catheter and drain

KEY POINT

Perforation can follow adequate resection of advanced tumour. It **rules out immediate intravesical chemotherapy**, because the drug would leak outside the bladder.

Post-operative intravesical agents

- P3 Immediate post TURBT intravesical Therapy for TCC — *rationale, agents, contraindications*

Not in the paper answer:

- Single mitomycin C or epirubicin **within 24 h** gives a ~**13%** absolute fall in recurrence (meta-analysis).
- Gemcitabine 2 g in S0337 caused no grade 4–5 toxicity.
- **BCG (bacille Calmette–Guérin) is never given immediately**: wait **2–4 weeks** with no gross haematuria († risk of systemic absorption).
- They reduce **recurrence**, not **progression**; give only without perforation. Gemcitabine (SWOG S0337) cut recurrence 12% at 4 years.

3 · Radical cystectomy: preparation

Why this matters: these steps cut ileus, infection and clots after RC.

- **Mark a stoma for every patient** (sitting and standing) and consent for a conduit.
- **Avoid routine mechanical bowel prep** and oral antibiotics; omitting them speeds recovery and cuts wound complications.
- **Alvimopan** (peripheral μ-opioid antagonist), given 30–90 min pre-op and then twice daily, lessens ileus and shortens stay.
- Give an IV antibiotic within 1 h covering skin and bowel flora, e.g. **cefoxitin**.
- Start **venous thromboembolism (VTE) prophylaxis** (compression and drug) **before induction** and extend it to **4 weeks**: 28 vs 8 days gives fewer major VTE, no extra bleeding.

Enoxaparin is standard; apixaban is equal, with better compliance.

- Men lie supine, flexed ~15°; women in low lithotomy. Prep with 10% povidone-iodine, not chlorhexidine, on the genitalia.

4 · Open radical cystectomy: technique

Why this matters: step order and sparing options are viva favourites.

ALGORITHM

1. Lower midline; Retzius space; divide urachus; incise peritoneum lateral to medial umbilical ligaments; divide vas or round ligament.
2. Mobilise caecum, small-bowel root and left Toldt line; make the **retrosigmoid window** (left ureter).
3. Free ureters with adventitia; divide obliterated umbilical artery for length; divide ureters.
4. Retractor (check vena cava); PLND; staple, clip or seal vesical pedicles.

- **Distal ureteric frozen section** is controversial: a positive margin predicts upper-tract recurrence, but clearing it neither prevents it nor proves survival benefit.

Male

ALGORITHM

1. Incise peritoneum over seminal vesicles; separate rectum; incise Denonvilliers fascia.
2. Control posterior pedicles; open endopelvic fascia.
3. Ligate dorsal venous complex (DVC); divide urethra.

- After radiation or fibrosis, use **sharp dissection under vision**. **Rectal injury is repaired primarily**, with or without a flap or faecal diversion.
- **Urethral frozen section** before a neobladder has a very high negative predictive value (prostatic urethral biopsy predicts poorly). **A positive result rules out orthotopic neobladder**; warn beforehand.
- **Nerve sparing**: few men are potent at 1 year (age predicts), and local recurrence occurs even in pT2. Avoid it in advanced disease or pre-existing erectile dysfunction.
- **Prostate sparing** improves potency and continence, but occult prostate cancer is common and distant metastasis frequent. It needs normal DRE and prostate-specific antigen (PSA), a TURP and a negative frozen section.

Female

- The standard is **anterior exenteration**, essential if EUA is positive or cT4a. Women present later, and female sex independently predicts cancer death.
- Counsel on sexual morbidity. Ovaries make testosterone after menopause, so discuss **ovarian sparing** if disease is not locally advanced.

ALGORITHM

1. Ligate ovarian vessels; incise posterior fornix; control pedicles.
2. Excise urethra from above or the introitus.
3. **Clamshell vaginal closure** (keeps girth, loses length), watertight, 2-0 polyglactin; pack out day 2.

- Vaginal pedicles are sinusoidal, so haemostasis must be meticulous; leaky closure leaks peritoneal fluid per vaginam.

- **Vagina and urethra sparing for neobladder** needs ≤cT2, an uninvolved bladder neck and a negative urethral frozen section. Dissect on a pack or EEA (end-to-end anastomosis) sizer, divide pedicles only to the mid-lateral vaginal wall, and preserve the striated sphincter.

5 · Pelvic lymphadenectomy

Why this matters: node yield reflects surgical quality and drives staging.

Template	Boundaries	Evidence
Standard	Lateral genitofemoral nerve; medial internal iliac artery; distal Cooper ligament; proximal ureter crossing common iliac	Reference
Extended	Standard plus common iliac and presacral	Better staging; possible survival gain
Superextended	Up to inferior mesenteric artery (preaortic)	No benefit over extended

- Protect the obturator nerve and seal lymphatics. **≥10 nodes** removed goes with better survival (NMIBC at RC: >20).
- A randomised trial found **extended not superior to limited** for recurrence-free or cancer-specific survival. Its staging value is certain, because it finds extra-pelvic node-positive disease for adjuvant therapy.

6 · Partial cystectomy

Why this matters: it suits only a narrow group.

- Indications are a **solitary small MIBC without CIS** and **urachal adenocarcinoma** (en bloc umbilicus, urachus and dome with frozen margins).
- Oncology is as for RC: PLND, NAC if eligible, and consent for RC. Muscle-invasive recurrence needs salvage RC.

ALGORITHM

1. PLND; open the bladder away from tumour.
2. Full-thickness excision with fat and a 1–2 cm frozen-section margin; reimplant ureter if needed.
3. 2–3-layer watertight closure; warm water irrigation (against seeding); drain.
4. Cystogram day 7, then remove catheter.

7 · Robotic radical cystectomy (RARC)

Why this matters: technical lapses cause unsalvageable recurrence.

KEY POINT

No effective salvage exists: urine spillage, poor PLND or a positive soft-tissue margin leads to near-uniformly fatal recurrence, so robotics is judged on oncology first.

- Indications match open surgery. Trendelenburg and pneumoperitoneum strain heart and lungs; prior surgery and obesity are relative contraindications.
- The Xi side-docks from the right (older systems between the legs). Ports sit higher than for prostatectomy (extended PLND, intracorporeal diversion), with a **15-mm left assistant port** for stapling and bagging packets.

ALGORITHM

1. Release sigmoid; develop presacral plane; open cul-de-sac to prostate apex.
2. Open Retzius; control anterior pedicle; free ureters (**never grasp**).
3. **Extended PLND with ureters intact**.
4. Clip ureters, send margin, pass left ureter under sigmoid.
5. Posterior pedicles, DVC, urethra (clip proximally only if neobladder).

- PLND before ureteric division keeps urine output as a resuscitation guide.
- Bag each packet whole. The obturator nerve is **not** the posterior limit: go down to the hypogastric vein. In women an EEA sizer gives countertraction.

8 · Complications and post-operative care

Why this matters: bowel problems prolong stay most.

- Complications are common but few are major; gastrointestinal ones lead, then infection.
- For **ileus**, use alvimopan, neostigmine (on telemetry), early feeding and opioid-sparing analgesia, with **no routine nasogastric tube**. Add VTE prophylaxis, mobilisation, spirometry.
- Ureteric devascularisation causes ischaemic stricture (keep adventitia); a non-watertight closure causes vaginal leak or fistula. Diversion complications: *see Ch 133–135*.

9 · Landmark trials

Trial	Design	Result	Take-home
Grossman 2003	NAC and RC vs RC	Median survival 46 vs 77 months	Offer cisplatin NAC
SWOG S0337	Single gemcitabine vs saline after TUR	12% fewer recurrences at 4 years	Immediate gemcitabine for low grade
RAZOR (2018)	Non-inferiority RARC vs open; open diversion	2-year progression-free survival 72.3% vs 71.6%; less blood loss, shorter stay	Robotic oncologically equivalent
iROC	RARC with intracorporeal diversion vs open	More days alive and out of hospital at 90 days; fewer VTE, wound complications	Recovery advantage

133 Use of Intestinal Segments in Urinary Diversion

1 · Surgical anatomy

Why this matters: a diversion depends on its pedicle's blood supply.

- **Stomach**: right gastroepiploic (gastroduodenal, from hepatic) and left gastroepiploic (from splenic); the greater-curve arcade carries the pedicle.
- The graft is antrum or a fundal wedge; if the left gastroepiploic is atretic (occasionally), use the right. A fundal wedge must **never reach the pylorus or lesser curve**; antrectomy is closed as Billroth I. Thick seromuscular layer suits a submucosal ureteric tunnel.
- **Small bowel** is ~22 ft: the proximal 2/5 is jejunum (wider, single arcades, thin mesentery) and the distal 3/5 ileum (multiple arcades). Bowel >**8 cm** from a straight vessel may not survive, so clear ≤8 cm of mesentery at a segment end.
- **Avoid pelvic small bowel after radiotherapy**: the last 2 inches of terminal ileum (fixed) and the 5 ft starting ~6 ft from the ligament of Treitz.
- **Colon** is supplied by the superior mesenteric artery (SMA: ileocolic, right colic, middle colic) and inferior mesenteric artery (IMA: left colic, and sigmoid branches ending as the superior haemorrhoidal). The internal iliac gives the middle haemorrhoidal; all link through the **marginal artery of Drummond**.
- A right colic arising from the ileocolic lets the ascending colon reach the pelvis.

Watershed	Site	Between
Griffiths point	Splenic flexure	Middle colic and ascending left colic
Sudeck's point	Rectosigmoid	Last sigmoid and superior haemorrhoidal

> **KEY POINT**
> Watershed zones get the smallest end branches of two arteries, so they go ischaemic first in hypotension or embolism; site anastomoses to one side.

2 · Segment selection

Why this matters: each segment carries its own metabolic and bowel penalty.

- P2 P1 Use of intestinal segments in urology / Intestinal segments in urology — *segment table (electrolytes, pros, cons, when to choose), NH_4Cl mechanism, alkali therapy*

Not in the paper answer:

- Losing **ileum** causes B12, bile-salt and fat malabsorption (loss of the ileal brake). Losing the **ileocaecal valve** causes bacterial overgrowth, vitamin A/D deficiency, fast transit, and bicarbonate and fluid loss. Losing **colon** causes diarrhoea and dehydration.
- **B12**: liver stores last **3–5 yr** and clinical deficiency is rare; homocysteine and methylmalonic acid are the sensitive markers.
- Gastric-surgery sequelae: dumping, small-stomach and afferent loop syndromes, anaemia (early: retention, bleeding, pancreatitis, duodenal leak).
- Neobladder reach improves with caecal mobilisation, mesenteric Z-plasty and bed flexion.
- **Stomach** suits renal failure, short gut and dense adhesions; prefer fundus to antrum, and consider a long-term H2 blocker or proton-pump inhibitor (PPI).
- **Haematuria–dysuria syndrome**: spasm, pain, haematuria or dysuria without infection, and skin excoriation; intermittent; treat with H2 blockers. Bacteriuria is no lower than with ileum

or colon.

- Use **jejunum** only if nothing else works, taking the most distal part. Colon causes obstruction needing reoperation in ~5% (half the ileal rate).
- Give empirical parenteral B12 after 5–7 yr of ileal diversion.
- Warn that clam cystoplasty brings more bowel dysfunction (overactivity goes with irritable bowel).

3 · Bowel preparation and enhanced recovery

Why this matters: routine prep is gone for most, but not all, cystectomies.

- P3 Methods of preoperative bowel preparation — *mechanical lowers amount, antibiotics lower concentration; oral regimens and harms; IV prophylaxis most important; ERAS; who still needs full prep*

Not in the paper answer:

- Oral antibiotics cut bacterial concentration to ~10^2/g; the descending colon holds **$10^{10\text{-}12}$/g**. Kanamycin is the best single agent.
- Colorectal data have a caveat: urology opens long segments into a sterile tract.
- Postoperative ileus (POI) is the commonest cystectomy complication, commoner than after colectomy (deperitonealisation, urine exposure, long surgery).
- Alvimopan is limited to 15 doses over ≤7 days and to inpatients; analgesia is kept, as it barely enters the brain.
- ERAS does not raise complications or readmissions and improves opioid use, diet, intensive care time and quality of life.
- The **IV antibiotic is most important**: give it 1–2 h before incision, covering **anaerobes**. Authors add clear liquids, magnesium citrate and oral neomycin the day before.
- **Enhanced recovery after surgery (ERAS)** uses **no routine bowel prep**. Chewing gum and metoclopramide help.
- **Alvimopan** is a peripheral μ-opioid antagonist: 12 mg 30 min–5 h pre-op, then twice daily.
- Salvage or benign cystectomy (radiation, fistula, neurogenic) still needs full mechanical and antibiotic prep and a slow diet.

4 · Intestinal anastomoses

Why this matters: bowel complications cause most early morbidity and death after diversion.

1. **Get exposure** and wall off the field; check the **blood supply** (transilluminate, bleeding edges, no tension).
2. **Prevent spillage** with non-crushing clamps, and irrigate the segment.
3. Join **serosa to serosa, inverted** and watertight, with **no strangulating sutures** and the **mesentery aligned**.

- Breakdown follows ischaemia, faecal sepsis, a **drain on the anastomosis** or irradiated bowel (most lethal). Sutured equals stapled; stapled side-to-side ileoileal is commonest.
- Leak presents late (2nd week). Neobladder–vaginal fistula causes continuous incontinence; vaginal-sparing cystectomy reduces it.
- **Obstruction** occurs in ~10% with ileum or stomach and ~5% with colon; adhesions cause more than recurrent cancer. Prevent with unirradiated bowel, closed apertures, omentum and **reperitonealisation**; after total exenteration, use polyglactin mesh and omentum for the pelvic floor.

- **Ogilvie syndrome** is a dilated caecum; a water-soluble enema excludes obstruction. If the caecum is **>12 cm** or white count rises, give neostigmine or decompress by colonoscopy. Operate for ischaemia, peritonitis or failure.

5 · Ureterointestinal anastomoses

Why this matters: stricture is a lifelong, silent threat to the kidney.

- Mobilise minimally, keep the periadventitia, spatulate, and sew mucosa to mucosa. **Bring the bowel to the ureter**, retroperitonealise, and use soft stents (they cut leak from 3–5% to near zero).

Technique	Type	Key feature	Stricture	Leak
Leadbetter–Clarke	Nonrefluxing	Colon; tunnel under taenia (2.5–3 cm)	8–14%	2.5%
Goodwin	Nonrefluxing	Transcolonic tunnel made from inside (3–4 cm)	No reliable data	
Bricker	Refluxing end-to-side	Any segment; simple	4–22% (mean 6%)	~3%
Wallace	Refluxing end-to-end	Ureters joined together (side-by-side, Y, head-to-tail)	**~3% (lowest)**	~2%

- Wallace: spatulate 1.5–2 cm; easy endoscopic access; tumour blocking both ureters is exceedingly rare. The best antireflux options are Pagano in colon and Le Duc or serosal apposition in small bowel.

> **KEY POINT**
> Conduit pressure is damped by ureteric peristalsis. Reflux harms the kidney only with impaired peristalsis plus infection or obstruction. Antireflux tunnels stricture more and do not stop colonisation, so a stented refluxing anastomosis is the default; reflux also allows a loopogram to image the upper tracts.

- Urinary fistula at days 7–10 (3–9%) causes fibrosis, then stricture.
- **Stricture** follows ischaemia, leak, radiation or infection, so surveillance is lifelong. The classic site is the **left ureter under the IMA**.
- Open reanastomosis is most durable; endoincision works in ~50–60% at 2 yr; balloon is not durable. Endourology fails more if the stricture appears <1 yr post-op, is **≥1.5 cm**, or is left-sided. Metal stents are only for limited life expectancy.

6 · Conduit diversion

Why this matters: the ileal conduit is the default; colon conduits cover exceptions.

- The **ileal conduit** is simplest, with fewest early complications; avoid it in short bowel, inflammatory small-bowel disease and irradiated ileum.

ALGORITHM

1. Take **10–15 cm** of ileum, 10–15 cm from the ileocaecal valve; keep a major arcade with windows ~5 cm.
2. Restore continuity by ileoileostomy, with the segment caudal; close the mesenteric window; flush the segment.
3. Pass the left ureter behind the sigmoid mesentery; make stented anastomoses.
4. Fix the base retroperitoneally in the right lower quadrant (RLQ); make the stoma.

- A missed duplex ureter leaks intraperitoneally. Stomal bleeding occurs in ~10%; intractable bleeding suggests portal-hypertensive varices, treated by transjugular intrahepatic portosystemic shunt (TIPS) or embolisation.
- A **jejunal conduit** is taken 15–25 cm from Treitz with a left upper quadrant stoma. Use only if ileum and colon are unusable; not in nutritional bowel disorders.

	Transverse	Sigmoid	Ileocaecal
Best for	**Extensive pelvic irradiation**; intestinal pyelostomy	Exenteration with colostomy	Long ureteric replacement
Supply	Middle colic	Sigmoid vessels	Ileocolic
Avoid	—	Sigmoid disease; ligated hypogastrics with rectum in situ; pelvic irradiation	—

- Avoid colon conduits in inflammatory bowel disease (IBD) and chronic diarrhoea.
- An **ileal vesicostomy** suits neurogenic bladder, best with detrusor–sphincter dyssynergia (DSD), and is reversible.
- Remove stents at days 4–6; drain retroperitoneally, **never intraperitoneally**.

Complication (%)	Ileal conduit	Colon conduit	Continent
Bowel obstruction	10	4	0.8
Ureterointestinal obstruction	8	Antireflux 20	4
Acidosis needing treatment	16	5	Ileum 8; colon 27
Renal deterioration	18	15	—

A gastric conduit causes no acidosis (0%).

7 · Stomas

Why this matters: stoma problems are the commonest after diversion, mostly technical.

- Mark sitting and supine; go **through the rectus** (lateral stomas herniate); stay ≥5 cm from the incision; avoid creases, scars, umbilicus, belt line and irradiated skin.
- For the aperture, **keep the subcutaneous fat**, cut the fascia in a cross, and make it admit two fingers.
- A **nipple (rosebud)** stoma suits an appliance (less stenosis, better fit): bring out 5–6 cm for a 2–3 cm spout; eversion sutures go to subdermis, never full-thickness skin. A **flush** stoma suits a continent catheterisable channel (low profile).

	End stoma	Loop (Turnbull)
Stenosis	Higher	Lower
Parastomal hernia	1–4%	4–20%

	End stoma	Loop (Turnbull)
Best for	Standard	**Obese** (thick, short mesentery)

- Hernia: ileal conduit 2–6.6%. Stenosis: ileal 20–24%, colon 10–20%, paediatric catheterisable >50%. Pseudoverrucous lesions respond to acidified urine and a fitted appliance.

8 · Renal function

Why this matters: continent diversion asks more of the kidney.

- Renal deterioration after conduits is 10–74% (ileal conduit 18%); ~6% die of renal failure. After ureterosigmoidostomy it is the leading cause of death beyond 15 yr.
- Continent diversions demand more than conduits (more bowel, longer dwell time). **Tubular acidification matters as much as glomerular filtration rate (GFR).**

Test	Threshold for continent diversion
Creatinine and urine protein	**<2.0 mg/dL**, no proteinuria
GFR	**>35 mL/min**
Ammonium chloride load	Urine pH **≤5.8**
Water deprivation	Osmolality **≥600 mOsm/kg**

- Exception: retentive diversion in children with end-stage renal failure awaiting transplant. **Bowel invalidates** urine osmolality and pH and serum urea and creatinine (reabsorbed).

9 · Metabolic complications

Why this matters: the segment predicts the electrolyte picture.

Absorption depends on the **segment, surface area, dwell time, solute concentration, renal function and pH**.

Segment	Na	K	Cl	pH	Associated	Treatment
Stomach	—	Low	Low	High	Raised aldosterone; lethargy, seizures, arrhythmia	H2 blocker, PPI; arginine HCl; remove segment
Jejunum	Low	High	Low	Low	Raised renin, angiotensin; vomiting, dehydration	IV saline, bicarbonate, thiazide
Ileum/colon	—	Low	High	Low	Total-body K depletion; hypocalcaemia	Alkali; chlorpromazine, nicotinic acid

KEY POINT
Stomach secretes HCl (alkalosis, low Cl); jejunum secretes NaCl (low Na and Cl, high K); ileum and colon absorb NH_4Cl (high Cl). **Only jejunum is hyperkalaemic**.

- **Gastric alkalosis** is usually trivial, but severe with renal failure or dehydration, as vomiting raises aldosterone and the kidney retains bicarbonate. Above a **gastrin** of **120 ng/L**, small gastrin rises cause large bicarbonate rises, and distension stimulates gastrin. Treat stepwise: rehydration and H2 blocker, arginine HCl if life-threatening, omeprazole, then segment removal.
- In the **jejunal syndrome**, NaCl loss causes hypovolaemia and switches on renin–aldosterone; the kidney makes low-Na, high-K urine, so the cycle continues. It is worse with proximal segments and parenteral nutrition, can be fatal, and needs long-term oral salt and thiazide.

- **Hyperchloraemic acidosis** is worst with **ureterosigmoidostomy** and colonic pouches (most Mainz patients need alkali). NH_4^+ also enters via K^+ channels. Villous atrophy occurs but transporters persist, so risk persists.
- Alkali: sodium bicarbonate (gas); Bicitra or Shohl's (poor taste); Polycitra when K is needed and Na unwanted. **Chlorpromazine 25–50 mg tds** (tardive dyskinesia) or **nicotinic acid 400 mg 3–4×/day** (avoid in ulcer or liver disease) inhibit cyclic AMP; they limit, not correct, acidosis.
- **Hypokalaemia** is commoner with colon, because ileum reabsorbs K. **Trap**: correcting acidosis alone causes profound hypokalaemia and flaccid paralysis, so give K with bicarbonate. Mg and Ca are also low.
- **Altered sensorium** is usually **ammonia** (liver dysfunction plus ureterosigmoidostomy), else low Mg or drugs. Treat: drain diversion, oral neomycin, protein restriction, IV arginine glutamate 50 g, lactulose.
- **Drugs are reabsorbed**, notably **phenytoin, some antibiotics and methotrexate**, so **drain continent pouches during chemotherapy**.
- **Osteomalacia** is commonest with ureterosigmoidostomy. Causes: acidosis (bone buffers H^+), vitamin D resistance, renal Ca loss (sulphate) and parathyroid hormone resistance. Alkaline phosphatase is raised. Correcting a base deficit >2.5 mEq/L prevents bone loss; treat with alkali and calcium, then active vitamin D.
- **Infection**: ~3/4 of conduit urines are bacteriuric; **treat pure Proteus or Pseudomonas** and observe asymptomatic mixed cultures.
- **Stones** (struvite, Ca phosphate) follow acidosis, urease organisms, **staples** and mucus: colon conduit 3–4%, ileal conduit 10–12%, continent caecal reservoir ~20%.

Cancer

- P4 Secondary tumours after urinary diversion — *Kälble risk table by diversion, surveillance, pathogenesis, adenocarcinoma at ureterocolic junction*

Not in the paper answer:

- The long-term ureterosigmoidostomy series average a mean of 11%.
- After **ureterosigmoidostomy**, cancer develops in 6–29% after **10–20 yr**. Defunctioned anastomoses still develop cancer, so **excise them, do not ligate**.

10 · Neuromechanical aspects

Why this matters: it explains why reservoirs are detubularised.

- † Detubularising bowel into a sphere disrupts coordinated peristalsis and, by Laplace's law, increases volume and lowers pressure. This is the basis of reservoirs and neobladders.

11 · Landmark trials

Trial	Design	Result	Take-home
Alvimopan (Lee 2014)	Randomised double-blind vs placebo, radical cystectomy	Bowel recovery 5.5 vs 6.8 days; stay 7.4 vs 10.1 days; less POI	Use with opioid-based analgesia

134 Continent Urinary Diversion

1 · Definition and basic principles

Why this matters: every continent design obeys the same three rules.

- P4 P2 Continent urinary diversion: orthotopic vs non-orthotopic / Principles of continence in continent diversion — *definition, three principles (reliable outlet, low pressure by detubularisation + sphere/Laplace, capacity ≥500 mL; 44 cm ileum <200 to 500–600 mL in a year), ileum stretches more than colon*

Not in the paper answer:

- Orthotopic is now preferred (natural voiding, no appliance, fewer revisions); cutaneous suits pre-existing incontinence or urethral disease.
- A large initial volume is unnecessary, as the reservoir stretches within a year.
- Detubularisation cuts the circular muscle, abolishing coordinated pressure waves; shapes include Kock double-folding, S and W.

KEY POINT

Continence is a pressure balance: low reservoir pressure (from detubularisation) against adequate outlet resistance (sphincter or efferent valve). If either fails, the patient leaks. Camey I (tubular ileum) failed because its pressure was too high.

2 · Patient selection

Why this matters: most are eligible, so selection means excluding a bad urethra, kidneys or hands.

- P4 Continent urinary diversion: orthotopic neobladder (Studer) versus non-orthotopic diversion — *eligibility 80–90% men / ~75% women, counsel all options (AUA/EAU), absolute and relative contraindications table, renal thresholds, ~10% intraoperative change, age/obesity not bars*

Not in the paper answer:

- **Locally advanced or node-positive (N+) disease found at surgery is not a contraindication**: pelvic recurrence is ~7% (13% if extravesical or N+) and rarely disables the neobladder. Exception: grossly positive margin at pubis, rectum or side wall. Adjuvant radiotherapy (RT) 45–50.4 Gy is well tolerated.
- eGFR 40–59 is safe (decline no faster than with a conduit); obstructive uraemia often reverses.
- A short or fatty mesentery is the commonest technical reason to abandon orthotopic diversion.
- Persistent prostatic urethral disease needs cutaneous diversion and urethrectomy. Keep the female urethra whatever the diversion (distal urethrectomy devascularises the clitoris).

ALGORITHM

1. Estimated glomerular filtration rate (eGFR) >40.
2. Intact external sphincter and adequate bowel.
3. Not frail; no severe liver, cardiac or psychological disease.
4. Men: no distal prostatic urethral cancer. Women: no palpable anterior vaginal tumour or cervical invasion.
5. If all are met and the patient will self-catheterise, offer an orthotopic neobladder.

- Mark a conduit site in everyone: crease-free standing, sitting and lying, away from umbilicus and scars, as lateral as possible yet through mid-rectus. Authors prefer the umbilicus for a continent cutaneous stoma.

Urethral recurrence

Feature	Men	Women
Dominant risk	Prostatic urethral involvement; worst with **stromal invasion** (usually N+, so give neoadjuvant chemotherapy)	**Bladder-neck tumour**; anterior vaginal wall invasion
Pre-op	Transurethral resection (TUR) biopsy at 5 and 7 o'clock beside the verumontanum — optional	Examination under anaesthesia (EUA); bladder-neck biopsy
Definitive test	Intraoperative **frozen section of urethral margin** (more predictive than TUR biopsy; the American Urological Association (AUA) requires a negative margin)	
Bladder carcinoma in situ (CIS)	Not predictive	
Skip lesions	—	None: a negative bladder neck means a negative urethra

- Risk rises with depth: none < CIS/mucosal/ductal < stromal. **The neobladder itself is protective** (hazard ratio (HR) 0.44).
- TUR misses much prostatic disease, so it is not repeated for this; authors send a urethral margin from the final specimen.
- Prostatic involvement needs urethral surveillance (wash cytology, urethroscopy).
- In women, vaginal invasion always accompanies bladder-neck disease; keep the urethra if bladder-neck biopsy and proximal urethral frozen section are negative.

Patient-related factors

- **Renal function** is the key contraindication. Proceed if creatinine < **1.7–2.2 mg/dL** or eGFR > **35–40 mL/min**. All need hydration and oral bicarbonate, avoiding nephrotoxins *see Ch 133*.
- To gain reach: deeper distal mesenteric division, discard a proximal segment, less table flexion and Trendelenburg, gentle traction.
- **Self-catheterisation** is needed by 10–20% of men and ~30% of women with a neobladder, and by all with cutaneous pouches. The need may appear years later, so all must be willing. Multiple sclerosis, quadriplegia or frailty favour a conduit.
- **Sphincter weakness**: artificial urinary sphincter (AUS) in men; Burch or sling in women (who will likely catheterise).
- **Prior radiation** is acceptable without incontinence, stricture, fistula or severe bowel damage. Brachytherapy, high-dose RT and cervical implants damage the rhabdosphincter most; the final decision is made intraoperatively.
- **Prior prostatectomy**: pre-op continence predicts outcome; flexible cystoscopy locates the old anastomosis. **Prior bowel resection**: reuse the old anastomosis as a reservoir end; short bowel means a sigmoid neobladder.

3 · Preserving continence at cystectomy

Why this matters: a neobladder is only as continent as the sphincter left.

The **rhabdosphincter** (pudendal nerve) is critical. It lies **anterior and lateral** to the urethra in both sexes.

Men

- Never dissect posteriorly beyond the apex. Incise endopelvic fascia and puboprostatic ligaments minimally; control the dorsal venous complex (DVC) with **absorbable** suture; clips or staples erode into the anastomosis.

> **ALGORITHM**
> 1. Open the endopelvic fascia; incise the puboprostatic ligaments slightly.
> 2. Ligate the DVC close to the apex.
> 3. Incise the anterior urethra and place sutures.
> 4. Place posterior sutures through the rectourethralis, then divide the posterior urethra.
> 5. Tag the sutures until the ureteroenteric anastomosis is done.

- Place sutures **before** full division, as the male urethra retracts. Apical frozen section with tumour means urethrectomy and cutaneous diversion or conduit; atypia or focal CIS means a second distal margin.
- Nerve sparing has a controversial continence benefit; it keeps membranous urethral sensation, which triggers reflex sphincter closure (night continence).

Women

- Sparing uterus and anterior vagina in selected low-stage disease avoids fistula, improves sexual function and reduces retention. Keep vaginal dissection no more dorsal than 2 and 10 o'clock. A palpable bladder-neck or vaginal tumour needs en bloc urethrectomy and cutaneous diversion.

> **ALGORITHM**
> 1. Divide the posterior pedicles and develop the plane off the anterior vagina.
> 2. Use the Foley balloon to mark the bladder–urethra junction.
> 3. Free the bladder neck and the first 0.5 cm of urethra.
> 4. Clamp the bladder neck, divide just distal to it, and send a frozen section.

- Leave endopelvic fascia untouched; the bladder neck is not needed for female continence. The female urethra does not retract, so suture after removal.
- Always interpose an **omental flap** between vagina and pouch; authors add sacrocolpopexy. If vaginal wall is excised, keep vaginal and urethral suture lines apart.

4 · Choice of bowel and ureteric reflux

Why this matters: ileum gives lowest pressure; obstruction, not reflux, loses kidneys.

- Distensibility: ileal circular, colonic circular, ileal longitudinal, then colonic longitudinal (least). **Ileum is the default**; its mucosa atrophies, lowering mucus and absorption. Colon has pressure waves; stomach has poor compliance and haematuria–dysuria.
- Avoid deep proximal mesenteric incision (reservoir supply). Close with absorbable suture; staples cause stones.
- **Antireflux is unnecessary**: Valsalva raises pressure equally throughout, and every valve can obstruct. A direct (Wallace or Leadbetter) anastomosis strictures in **3–6%**, against 7–29% for Le Duc. **Obstruction, not reflux, destroys kidneys.** Kock nipple failures are late (>10 yr) and silent until bilateral hydronephrosis.

5 · Orthotopic neobladder designs

Why this matters: Studer and Hautmann dominate.

- P2 P4 Use of intestinal segments in urology, and orthotopic neobladder — *design table (Studer, Hautmann, T pouch, Abol-Enein, Camey II, Hemi-Kock, Padua, Mainz III, sigmoid); Studer steps; Hautmann vs Studer*

Not in the paper answer:

- Hautmann became refluxing in 1997, cutting stenosis from 9.5% to 1%; a 40 cm version limits retention and reabsorption; a 5 cm U-flap sits at the urethral point.
- Hemi-Kock has been abandoned. The sigmoid neobladder has the worst night continence.
- In the **T pouch**, the distal 3–4 cm of the afferent limb lies in a serosal trough, fixed through mesenteric windows and covered by ileal flaps. Tapering the limb was abandoned (ostial stenosis).
- Studer is the most popular design. Padua (VIP) is used robotically.

ALGORITHM

Studer:

1. Isolate ileum ending 15–20 cm proximal to the ileocaecal valve; discard ~5 cm proximally; keep a 40–44 cm reservoir (two 20–22 cm limbs) and an afferent limb.
2. Open the reservoir, oversew the back wall, and fold it the opposite way.
3. Make stented end-to-side ureteroileal anastomoses.
4. Join the dependent point to the urethra.

KEY POINT

Hautmann vs Studer is a capacity trade-off: a bigger initial pouch is continent sooner but empties worse and absorbs more later.

6 · Continent cutaneous reservoirs

Why this matters: without a urethra, all depends on a durable, catheterisable valve.

- P2 Principles of continence in continent urinary diversion; Mitrofanoff principle — *nipple vs flap vs imbricated valve, appendix ≥5–6 cm/≥12 Fr, Mitchell extension, Yang–Monti, tapered ileum, Indiana*

Not in the paper answer:

- An **umbilical stoma** has less stenosis, is easiest for paraplegics and is near-invisible (so wear a medical alert bracelet). A lower-quadrant stoma is concealed by clothing. The appendix takes only 14–16 Fr catheters.
- **Indiana** uses the right colon and ~10 cm of terminal ileum. The valve is double-imbricated with non-absorbable Lembert sutures over 3–4 cm on a 16 Fr catheter, and the ileum is tapered. The whole colon is opened, because residual caecal peristalsis overcame the valve.
- Nipple valves (Kock) are historical.
- The authors' **right colon pouch** uses ~30 cm colon, with intact terminal ileum as afferent limb (ileocaecal valve as antireflux). The appendix lies in a 4–5 cm taenial trough to the umbilicus; if unsuitable, use Yang–Monti.
- Intra-op, fill the pouch, check continence and catheterisation, and fix it to the abdominal wall.

7 · Perioperative care

Why this matters: ileus and mucus blockage are the early problems.

- **Enhanced recovery after surgery (ERAS)** means prehabilitation, no oral mechanical prep (the authors still prepare colon), carbohydrate loading, restrictive fluids, no nasogastric (NG)

tube, early diet and opioid-sparing analgesia. **Alvimopan** cut ileus from ~23% to ~7%. Ileus is the commonest gastrointestinal complication; up to 70% are malnourished or frail.

- **Drainage**: 7 Fr single-J stents and a 24 Fr urethral catheter irrigated for mucus; remove the catheter at **3 weeks**; pouchogram only if drain output is significant. Then timed voids, lengthened progressively, with pelvic floor exercises. For robotic diversion *see Ch 135*.

8 · Results and complications

Why this matters: men tend to leak, women to retain.

Early (<90 days) morbidity is no higher than with an ileal conduit; colonic cutaneous pouches have more infection and dehiscence.

Complication	Key facts	Management
Urine leak	Commoner (long suture lines)	Catheter, stents; if undrained, percutaneous drain with or without nephrostomies; **avoid early open repair**
Bacteriuria	Very common	Treat only if symptomatic or *Proteus/Pseudomonas*; in febrile urinary tract infection (UTI), look for obstruction, stones, residual; if recurrent, nitrofurantoin or methenamine
Ureteroileal stricture	Can appear after 10 yr, often silent	Suspect on CT or ultrasound (US); confirm by pouchogram or antegrade study; endoscopy (retreatment >50%) inferior to open
Pouch stones	Kock and stapled pouches	Endoscopic
Pouch perforation	Rare; radiation, chronic retention	Cystogram; catheter with or without drains if stable
Pouch–vaginal fistula	5–10%; vaginal excision, radiation	Lateral cystogram before catheter removal; beyond the first weeks it won't heal, so repair transvaginally or transabdominally, or convert to cutaneous
Stomal stenosis (cutaneous)	Up to ~20%; after 90 days, radiation	Usually open revision
Efferent limb necrosis	Rare, ischaemic	Endoscopy for level; explore if below fascia
Incisional hernia	Common; impairs emptying	Repair

Continence

- Day continence comes over **3–6 months**, ultimately **80–90%**. Night continence lags into year 2, and nocturnal incontinence persists in **20–50%**. Risks: age >65, colon, diabetes, no nerve sparing, hysterectomy.
- Night leakage reflects no sphincter–detrusor reflex, lower nocturnal tone and bowel water secretion; void at bedtime and set an alarm 2–3 times.
- Evaluate after the 12–18-month plateau: exclude fistula, use biofeedback, urodynamics for pressure waves (colon).

Stress incontinence	Men	Women
Bulking agent	Marginal	Mild only; <50%, poorly durable
Definitive	AUS or male sling	Pubovaginal sling — most then catheterise
Safety rule	No blind retropubic needle passage; use bone anchors or the prepubic route	

- A persistent cutaneous stomal leak needs surgical revision of the efferent mechanism; a minor one, bulking.

Retention

- It presents as infection, overflow, palpable pouch or new hydronephrosis. It affects men in 4–10% (~21% by 10 yr; long bowel, prostate sparing) and **women in 20–30%** (up to ~60% self-catheterise on questionnaires).
- In women the cause is **mechanical**: the pouch falls back and angulates the pouch–urethral junction. Prevent with omentum behind the pouch, vaginal suspension, sacrocolpopexy and uterine preservation.
- Treat with clean intermittent self-catheterisation (CISC); drugs are useless. Exclude stricture (~1%) or recurrence; incise an obstructing mucosal flap endoscopically.

> **KEY POINT**
> Men fail by **leaking** (sphincter without reflexes), so augment the sphincter. Women fail by **not emptying** (unsupported pouch kinks), so support the pouch.

9 · Follow-up

Why this matters: late complications are silent, so follow-up is lifelong.

- Phases: stricture in the first 4 months; recurrence from 4 months to 3 yr; stones, late obstruction and urethral or upper-tract tumour after 3 yr.
- Review 4-monthly in year 1, 6-monthly to 3 yr, then annually (examination, metabolic panel, blood count), with CT and chest imaging to 5 yr.
- Check B12 yearly. Do urethral wash cytology in men (especially prostatic CIS), and bimanual examination and voided cytology in women (urethral recurrence is invisible on CT). After 5 yr continue lifelong upper-tract US, metabolic panel and blood count.
- Quality of life (QoL): no diversion is proven superior; the neobladder gives better body image and the conduit better urinary function.

10 · Landmark trials

Trial	Design	Result	Take-home
USC Studer vs T pouch	Randomised, 484 radical cystectomy (RC) patients	No difference in renal function or infection at 3 yr; T pouch more secondary procedures	Antireflux adds cost without benefit
RAZOR	Open vs robotic RC, extracorporeal diversion	Continent diversion only 21%	Low uptake even with expert surgeons
iROC	Open vs robotic RC, intracorporeal diversion	Continent diversion 11%	Intracorporeal approach further reduces continent diversion

135 Minimally Invasive Urinary Diversion

1 · Context

Why this matters: robotic cystectomy evidence does not cover diversion built inside.

- Robot-assisted radical cystectomy (RARC) equals open oncologically in trials (RAZOR), but with **extracorporeal** diversion (ECUD). Intracorporeal diversion (ICUD) is untested.
- In the International Robotic Cystectomy Consortium (IRCC), ICUD rose ≈ **11%/year**, driven by the **ileal conduit** (commonest robotic diversion).
- Continent reservoirs declined before robotics; no good evidence shows better quality of life (QoL).

2 · Patient selection

Why this matters: a bar to one step is not a bar to all.

High body mass index, age, locally advanced disease and prior surgery no longer exclude. Three separate decisions:

Decision	Contraindication
Minimally invasive route	Poor cardiopulmonary tolerance of steep Trendelenburg (relative; eased by Xi and AirSeal)
Use of ileum	Short bowel; severe inflammatory bowel disease; radiation enteritis
Neobladder (absolute)	Urethral involvement distal to prostate; creatinine **>2 mg/dL**; poor hepatic function (as for open)

ALGORITHM

Pasadena Consensus learning pathway:

1. RARC with extracorporeal diversion.
2. Intracorporeal conduit (≈30–100 cases).
3. Continent diversion (≈25–50 cases).

Target time: **≤4 h conduit, ≤5 h neobladder**.

3 · Perioperative bundle

Why this matters: ileus delays recovery most.

Enhanced recovery (ERAS) uses alvimopan, goal-directed fluids, immediate oral diet and opioid-sparing analgesia (paracetamol or ketorolac, not epidural or patient-controlled analgesia). This gives less ileus and shorter stay. Caution: a stay **<5 days** about doubles 90-day major complications.

Element	Recommendation
Fasting	Solids to 6 h, clear fluids to 2 h
Bowel preparation	**None**, no oral antibiotics (electrolyte upset; liquid spillage; no benefit)
Thromboprophylaxis	Pneumatic compression and stockings; low-molecular-weight heparin before and up to **4 weeks** after
Antibiotics	Broad-spectrum intravenous within **1 h**

4 · Position and ports

Why this matters: bowel work is high, so ports sit higher.

- Use lithotomy and steep Trendelenburg (the Xi table allows flatter), Hasson entry after prior surgery, and a 0° lens (30° down for a deep female pelvis).
- **Six to seven ports sit more cephalad** to reach over small bowel and take nodes to the aortic bifurcation.
- Camera 12 mm above umbilicus; 8-mm arms lateral to rectus; fourth arm above and medial to the anterior superior iliac spine.
- A neobladder adds a 12–15-mm left stapling port. A suprapubic 12-mm port serves bowel reanastomosis and extends to a Pfannenstiel for the specimen.
- Undock and reduce Trendelenburg for the urethra–neobladder anastomosis.

5 · Nerve sparing

Why this matters: one autonomic plexus serves sexual function and neobladder continence, so nerve sparing is a continence operation.

Select a safe tumour, low burden and good baseline function.

	Male	Female
Danger points	In order: presacral and medial common iliac; high pedicle ligation; seminal vesicles and Denonvilliers fascia; prostatic pedicles and apex	Lateral vaginal wall and bladder base (inferior hypogastric plexus)
Manoeuvre	Athermal, intrafascial, traction-free pedicle control	Preserve tissue around proximal urethra and anterior vaginal wall; extract via Pfannenstiel, not transvaginally
Sphincter	One-third overlapped by dorsal venous complex (DVC); dorsal penile nerve branches within **3 mm** of apex	Mainly mid–distal third, so continence survives loss of bladder neck

6 · Intracorporeal ileal conduit

Why this matters: the conduit is the usual first intracorporeal step.

ALGORITHM

1. Pass the left ureter under the sigmoid mesentery to the right.
2. Choose 12 cm ileum, 15–20 cm from the ileocaecal valve; place the marionette stitch.
3. Isolate with a 45-mm stapler.
4. Do ureteroileal anastomoses (left first) and stent.
5. Reanastomose bowel, then form the stoma.

- The **marionette stitch** (1-0 silk through hypogastrium and stoma end) is held **untied**. It replaces the missing hand and positions the conduit.
- Avoid a redundant segment *See Ch 133*. Tag cut ends to prevent malrotation; washout is abandoned.
- The **enterotomy** is an athermal cold-scissors buttonhole with serosal removal, because thermal slits gave more strictures. Retroperitonealise the anastomosis.
- Stents are 8.5-Fr single-J. The **guidewire comes out only after anchoring**; stent ends exit via the 15-mm port.

- Restore bowel after both ureters. Use Bricker (separate end-to-side) or Wallace (Y, one anastomosis).
- For the stoma, a clamp through the site grasps marionette and stents; undock, deliver the conduit and check orientation.

7 · Intracorporeal neobladder

Why this matters: the order of steps differs from open.

The key change is **early neobladder–urethral anastomosis** (Karolinska): dependent ileum is anchored to the urethra first, then the pouch is built in place. Design principles: *See Ch 134*.

Design	Ileum	Signature
Karolinska (modified Studer)	50 cm	10-cm afferent limb; Penrose sling
USC (Studer)	65 cm (longest)	44+16 chimney+5 discarded
UCLH Pyramid	50 cm	Koie-derived cross-fold
Saint-Augustin modified Y	40 cm (shortest)	Refluxing, uncrossed ureters
Padua	42 cm	Only stapled pouch; split-nipple antireflux
FloRIN	50 cm	Asymmetrical U; L-shaped neo-trigone; antireflux
Roswell Park (Hautmann W)	45–50 cm	One trough at a time; chimneys

ALGORITHM

1. Urethral anastomosis (Van Velthoven barbed suture).
2. Detubularise distal 40 cm (cold scissors); form the posterior plate.
3. Join ureters to afferent limb or chimney, stented.
4. Close the anterior wall once the 22-Fr catheter is confirmed.

- Bowel is restored via the suprapubic port (60-mm side-to-side stapled); close the mesenteric window.
- A **tension-free urethral anastomosis** lacks tactile feedback. Help comes from maximal urethral length, sigmoid out of pelvis, less Trendelenburg, perineal pressure, barbed suture, Penrose traction (Karolinska) and a Rocco-type stitch (USC). Incising mesenteric peritoneum risks pouch ischaemia.

Intracorporeal Indiana pouch

ALGORITHM

1. Mobilise the right colon.
2. Divide ileum 15 cm from the ileocaecal junction and transverse colon proximal to the middle colic; restore ileocolic continuity at once.
3. Detubularise along the anterior taenia and fold.
4. Bring ureters through the posterior plate on double-J; 24-Fr caecostomy via appendiceal stump.
5. Taper ileum over 14 Fr, imbricate the valve, and form an umbilical stoma.

KEY POINT

Capacity comes from detubularised right colon; continence from tapered ileum and the imbricated ileocaecal valve.

8 · Postoperative care

Why this matters: each tube has its own removal time.

Avoid overhydration; no routine nasogastric tube (Cochrane). Irrigate the neobladder **8-hourly** for mucus.

Tube	Removal
Urethral Foley as pelvic drain (conduit)	Day 1
Neobladder catheter	**3–4 weeks**, after leak-free pouchogram
Externalised stents / double-J	≈1 week / cystoscopically at 3–4 weeks
Feeding-tube stents tied to Foley	With the catheter

9 · Outcomes

Why this matters: most problems come from the diversion, and many appear late.

- **Operative time** is a quality measure. Diversion type determines it most (conduit ≈70 min shorter); ICUD and ECUD take similar time. Proficiency needs ≥**66 RARCs**.
- **Complications** are mostly diversion-related, infection commonest. Two-thirds of reoperations occur after 3 months, so 90-day reporting underestimates. A trial meta-analysis found no difference in 90-day major complications.
- **Parastomal hernia** reaches ≈one-third by 3 years (risks: long operation, fascial defect ≥30 mm, low estimated glomerular filtration rate). Robotic mesh repair is durable.
- ICUD versus ECUD (IRCC): after adjustment, neutral for high-grade complications and readmission; less transfusion.
- **Reoperation** reaches ≈half by 5 years; predictors are neoadjuvant chemotherapy and any high-grade complication.
- **Uretero-enteric stricture** rises over 5 years. Robotic revision equals open; indocyanine green (ICG) localises the ureter.

> **KEY POINT**
> Night continence is worse because the urethrosphincteric guarding reflex is inactive in deep sleep. Continence then depends on reservoir capacity, hence detubularisation, nerve and sphincter preservation, a night alarm and limiting night output. Assess at 6–12 months.

- Good Eastern Cooperative Oncology Group (ECOG) status and retained filling sensation predict continence. Up to 25% of men need clean intermittent catheterisation (CIC) after open neobladder.
- Intracorporeal series: day continence 68–100%, night 55–85%, CIC up to 11%. Emptying and night continence lag; only FloRIN matched open on urodynamics.
- QoL: physical domains recover by 24 months, except sexual function.

10 · Landmark trials

Why this matters: RAZOR and iROC answer different questions.

Trial	Design	Result	Take-home
RAZOR	RCT RARC vs open, extracorporeal diversion	Oncological equivalence	Says nothing about ICUD

Trial	Design	Result	Take-home
iROC	Multicentre RCT, intracorporeal vs open	Open: worse QoL at 5 weeks, more disability at 5 and 12 weeks; no difference after 12 weeks; 11% continent	ICUD gains are early and recovery-related

124 Bladder Surgery for Benign Disease

1 · Surgical anatomy essentials

Why this matters: anatomy decides whether surgery stays safe and extraperitoneal.

- The bladder is an extraperitoneal hollow organ (mucosa, lamina propria, detrusor, perivesical fat) arising from the urogenital sinus (endoderm).
- **Anteriorly** is the space of Retzius, the usual access; transversalis fascia separates bladder from abdominal wall. **Posteriorly**, peritoneum reflects near the prostatorectal junction (men) or vesicouterine space (women); Denonvilliers fascia separates bladder and prostate from rectum. **Laterally**: pelvic sidewall, obturator space, vas or round ligament.
- The **medial umbilical ligaments** are the obliterated umbilical arteries. The **median umbilical ligament** is the urachus (allantois remnant), which can persist as a patent urachus, sinus or cyst.
- The **uterine artery** crosses over the ureter near the internal os ("water under the bridge"); the **vas** crosses it near the seminal vesicle tip. The distal ureter runs medial to the medial umbilical ligament.
- The superior vesical (from umbilical) and inferior vesical (internal iliac; vaginal artery in women) arteries form the posterolateral pedicles. Lymph drains mainly to external iliac, also obturator, internal and common iliac nodes.
- Nerves: sympathetic T10–L2 and parasympathetic S2–S4. The pudendal nerve (Onuf's nucleus, S2–S4) supplies the urethral sphincter.

KEY POINT
Three landmarks organise every benign bladder operation: Retzius for access, the peritoneal reflection for the posterior bladder, and the distal ureter for safety. They decide whether surgery stays extraperitoneal.

KEY POINT
Common aftercare: watertight **two-layer 2-0 polyglactin** closure (mucosa, then seromuscular), leak test, large-bore Foley for about **1 week**, self-suction drain out at about **48 h** if output is low, and a **cystogram before catheter removal**. With high drain output, drain creatinine above serum means a urine leak.

2 · Bladder diverticula

Why this matters: a sac without muscle cannot empty, so stasis brings infection, stones and cancer.

- P2 P3 Bladder diverticulum / Hutch diverticulum — *definition (no muscularis propria), congenital vs acquired table, VCUG/cystoscopy/CT/urodynamics work-up, indications, tumour inside means cystectomy*

Not in the paper answer:

- Most acquired diverticula are asymptomatic and incidental.
- Voiding cystourethrogram (VCUG) shows diverticula hidden among trabeculae and bladder neck obstruction (hypertrophy, dyssynergia, valves, stricture). Ultrasound shows hydronephrosis but not the ureter or stones well.
- Intraperitoneal urine leak is a major source of morbidity, which is why open surgery stays extraperitoneal.

- A fibrous **pseudocapsule** gives the surgical plane. Other causes are posterior urethral valves, prune-belly and iatrogenic injury (women after anti-incontinence surgery). For a congenital (Hutch) sac, still screen for voiding dysfunction and plan reimplantation.
- At cystoscopy, enter every sac and note neck size, distance from the orifice, any intravesical lobe, and stricture or bladder neck contracture (treat these before or with the diverticulectomy). Remove stones at surgery.
- VCUG decides between a preoperative stent and simultaneous reimplantation. If contrast cannot be given, use ultrasound or non-contrast CT plus on-table retrograde studies.
- Computed tomography (CT) urogram shows hydronephrosis (from compression or vesicoureteral reflux, VUR), ureteric displacement and stones. Send cytology in all; add flow rate or urodynamics, prostate volume if treating benign prostatic hyperplasia (BPH), and a **negative urine culture** before surgery.

Situation	Management
Small, asymptomatic, uncomplicated	Observe, or endoscopic fulguration/resection
Large with residual urine, recurrent urinary tract infection (UTI), stones, pain	Diverticulectomy
VUR or ureteric obstruction	Diverticulectomy with or without reimplantation
Cancer in diverticulum	Partial or radical cystectomy, not diverticulectomy

Operative principles

- Treat the cause at the same sitting: transurethral prostate surgery, or simple prostatectomy for glands **>80 cc**, and augmentation for neurogenic bladder. Do **endoscopic outlet surgery first**, because later irrigation disrupts fresh suture lines.
- Localise the sac with a Councill-tip catheter over a wire for selective filling, or by flexible cystoscopic transillumination (best if multiple). Place a ureteral catheter if the sac is near the ureter.
- Open surgery is extraperitoneal even for posterior sacs and remains the gold standard. Laparoscopic and robotic surgery is usually transperitoneal and reasonable in experienced hands.

ALGORITHM

Open intravesical diverticulectomy

1. Put the anterior superior iliac spine (ASIS) at the table break; fill via an 18 Fr Foley with about 200 mL.
2. Lower midline incision; split recti; incise transversalis fascia.
3. Sweep peritoneum off the dome and open a vertical cystotomy with 2-0 stays.
4. Identify the orifices (5 Fr ureteral catheter if concerned) and excise the sac.
5. Place a 20–22 Fr Foley, close in two layers, leak-test; optional suprapubic catheter; pelvic drain.

Step	Intravesical	Combined intra-/extravesical
Delivery	Allis clamp everts the sac, or it is dissected from the pseudocapsule from inside; mucosal incision about **0.5 cm** round the neck	A finger in the sac pushes it out; the neck is incised circumferentially from outside
Hazard	Ureteric orifice injury	**Excess bladder wall resection** if the inner and outer incisions do not meet
Closure	Two layers 2-0 polyglactin, secure seromuscular layer (prevents recurrence); no drain in capsule	

Laparoscopic / robotic

- Use lithotomy, steep Trendelenburg, and cystoscopy first. Pneumoperitoneum is **15 mm Hg**, with the assistant port opposite the sac.
- Incise peritoneum over the sac (drop the bladder if large or multiple), circumscribe the neck, and excise and bag it with the ureter in view. Staplers risk stones forming on the staples.
- The transvesical variant opens the bladder and inverts the sac; be cautious if the ureter is involved. Close in two layers 2-0 polyglactin; peritoneum may be closed; drain via a trocar site.
- If the sac sits at the ureteral hiatus, amputate the neck, close the opening, make a new hiatus, place a double-J and reimplant.
- Alternatives for posterior sacs are extraperitoneal laparoscopy, pneumovesicoscopy, laparoendoscopic single-site surgery (LESS) and the single-port robot. The drain comes out at 24–48 h.

Complications

- **Urinary extravasation is commonest** (open and minimally invasive alike). It shows as high drain output or infected urinoma, and after transperitoneal surgery as ileus and urinary ascites. Check drain creatinine and do CT cystography.

ALGORITHM

Managing a leak

1. If asymptomatic, keep Foley and drain.
2. If a symptomatic collection forms, drain it percutaneously.
3. Do a cystogram before removal; if the leak persists, choose nephrostomies or exploration and closure.

- Unrecognised ureteric injury presents with acute kidney injury (AKI) or flank pain. Image, then try a retrograde stent; if it fails, nephrostomy and delayed repair.
- Others: UTI, fistula, bladder spasm, wound infection, bowel injury.

KEY POINT

Diverticulectomy is a dissection problem: find the sac, stay in the sac–pseudocapsule plane, keep the ureter in view, close the seromuscular defect securely, and fix the outlet first.

3 · Bladder stones, foreign bodies and fistulae

Why this matters: a bladder stone signals stasis, so treat the cause too.

- The cause is **stasis**: bladder outlet obstruction (BOO), neurogenic bladder, or augmented and substituted bladders. Foreign bodies act as a nidus and are mostly iatrogenic (mesh, migrated intrauterine device).

- Most are asymptomatic; others cause haematuria, irritative voiding, recurrent UTI or retention.
- **Non-contrast CT** is best for stone size and volume, prostate volume and hydronephrosis. Cystoscopy is needed only for haematuria, suspected cancer or prostate assessment.
- Treat the stone **and** its cause; the concomitant operation often picks the route (large prostate: transvesical prostatectomy with cystolithotomy; diverticulum: diverticulectomy; neurogenic: enterocystoplasty).

Approach	Best for	Advantage	Limitation
Transurethral cystolitholapaxy	Most stones	No incision	Prolonged instrumentation risks urethral stricture
Percutaneous cystolithotripsy	Large stones	Nephroscope, short stay, no stricture	Contraindicated after prior surgery, diversion or augmentation (bowel injury)
Open cystolithotomy	Very large burden; with prostatectomy, diverticulectomy, augmentation	Historic gold standard	Incision, longer catheter
Laparoscopic/robotic	Large burden with other procedures; foreign bodies, eroded mesh	Minimally invasive	Technically demanding

- **Percutaneous**: through a 1-cm incision two fingerbreadths above the symphysis, a needle is passed under ultrasound or cystoscopic vision and dilated to a **30 Fr** sheath for a 24–26 Fr nephroscope. Stones <1 cm are grasped whole, larger ones fragmented; the suprapubic catheter is clamped at 24–48 h.
- **Open**: via Pfannenstiel or midline, open the bladder vertically, remove stones, inspect for tumour and intravesical lobe, do concomitant surgery, close in two layers and drain. After prior surgery, image before percutaneous access to exclude interposed bowel.
- **Transvesical laparoscopy** uses a pneumovesicum of **8–12 mm Hg**, avoids the peritoneum and suits eroded mesh. Transperitoneal laparoscopic or robotic cystolithotomy allows simultaneous prostatectomy or diverticulectomy. Aftercare: Foley about 1 week, cystogram.
- Refractory fistulae (colovesical, vesicovaginal) may need partial cystectomy, and a refractory urinary fistula may justify simple cystectomy.

4 · Partial cystectomy for benign disease and urachal surgery

Why this matters: some benign lesions behave locally like cancer.

- Indications are leiomyoma, paraganglioma, endometriosis, urachal anomalies and fistulae. Most benign tumours need only transurethral resection (TUR).
- **Avoid it with small capacity or severe dysfunctional voiding.** Exclude malignancy first, because cancer needs wide margins and lymphadenectomy. Lesions near the orifice need reimplanting.

Lesion	Clue	Investigation	Treatment / trap
Leiomyoma	Women of childbearing age; endovesical, intramural or extravesical	Transvaginal ultrasound (US), MRI	Small endovesical: TUR; large or intramural: wide local resection

Lesion	Clue	Investigation	Treatment / trap
Paraganglioma	Episodic hypertension; **hypotension or syncope on micturition or sex**; or painless haematuria	Plasma metanephrines; MIBG scan; yellow raised mucosa	**Alpha-blockade before any instrumentation**; partial cystectomy preferred over TUR (recurrence)
Endometriosis	Cyclical haematuria with menses	US, MRI (bright on T1); cystoscopy limited	Partial cystectomy (disease is extravesical)
Urachal anomaly	Umbilical discharge, mucinuria, infected mass; malignant risk	US/CT; VCUG or sinogram; cystoscopy	Culture and antibiotics, with or without drainage; if recurrent, excise urachus with a dome cuff

- Dome and anterior lesions are approached extraperitoneally, posterior ones transperitoneally (mobilise the ureter; develop pubocervical or Denonvilliers fascia for base lesions). No node dissection. Localise by cystoscopy or ultrasound, and **aspirate before opening to avoid spillage**.
- Excise full thickness with about a **2-cm margin** if capacity allows, with frozen-section margins; reimplant if the orifice is involved; 22 Fr Foley, two-layer closure, drain. Minimally invasive: transillumination or laparoscopic ultrasound outlines the margin, and a ureteral catheter protects the ureter.
- **Urachal excision** takes the whole tract with peritoneal wings and a small dome cuff; **no wide margin or frozen section**; the umbilicus may be kept if benign. Open: periumbilical midline incision, circumscribe the umbilicus, excise down to a transmural dome cuff. Minimally invasive: camera 5 cm above and 2 cm left of the umbilicus; urachus dissected last.
- Complications: extravasation (commonest), ureteric injury, reduced capacity (counsel).

> **KEY POINT**
> Leiomyoma, paraganglioma and endometriosis infiltrate and recur if incompletely excised, so benign partial cystectomy borrows cancer discipline (exclude cancer, no spillage, a margin). Urachal surgery is the exception: the whole tract matters, not a wide margin.

5 · Simple (supratrigonal) cystectomy

Why this matters: a last resort that helps only if the bladder is truly the source.

- It treats end-stage refractory dysfunction: radiation cystitis, neurogenic bladder, refractory incontinence, fistula, bladder neck contracture, refractory anastomotic stricture, and interstitial cystitis/bladder pain syndrome (IC/BPS). The aim is to preserve renal function or quality of life.
- Prove the bladder is the source, since non-vesical pain persists. In IC/BPS, success is predicted by a **small fibrotic bladder, small capacity under anaesthesia and Hunner lesions**.
- Diversion alone risks pyocystis, bleeding, pain and secondary cancer; many IC/BPS patients later need cystectomy.

Feature	Simple (benign)	Radical (cancer)
Prostate / anterior vagina	Preserved unless indicated	Removed
Lymphadenectomy	None	Performed

Feature	Simple (benign)	Radical (cancer)
Bladder opened	Often, to find the trigone and bladder neck	Avoided
Excision	Bivalve and amputate near the trigone; stump oversewn and residual mucosa cauterised	En bloc

Steps: divide the ureters, fill and bivalve the bladder, divide pedicles and amputate each half, and cover the stump with posterior peritoneum; usually an ileal conduit follows. Care mirrors radical cystectomy; robotic surgery may reduce complications. Drains stay 5–7 days (out when drain creatinine equals serum). Complications (wound infection, UTI, transfusion, ileus) are common.

6 · Ureteral reimplantation in adults

Why this matters: adults need reimplantation mainly for injury and stricture, not reflux.

- Indications are distal ureteral trauma, iatrogenic injury, **stricture** and late-presenting injury (the commonest elective reasons), and reimplantation alongside diverticulectomy or partial cystectomy. Primary VUR is a rare indication.
- In thermal or gunshot injury, debride even normal-looking ureter; assess partial injury with on-table retrograde pyelogram or methylene blue.
- Strictures follow impacted stone, radiation or chronic inflammation. An atrophic kidney needs decompression, then a renogram. With a stent in place, do a retrograde study before removal.
- Measure the gap with antegrade and retrograde imaging; exclude malignancy (cytology, ureteroscopic biopsy, frozen section). A **poorly functioning kidney needs nephrectomy, not reimplantation**. **Persistent extravasation despite adequate stenting** is an indication.

> **KEY POINT**
> Refluxing reimplantation is acceptable in adults. VUR is rarely the indication, and refluxing anastomoses show no faster renal decline, more stenosis or more infection than tunnelled ones, while needing less ureter and less bladder mobilisation.

Feature	Nonrefluxing (tunnelled)	Refluxing (direct)
Tunnel	**3–5 × ureteric diameter** (5:1 robotically)	None
Ureter length / mobilisation	More	Less; reaches the posterior dome
Use	High-grade reflux with renal risk	Generally sufficient in adults

Technique	Access	Neo-hiatus	Signature step
Modified **Lich-Gregoir** (nonrefluxing)	Extravesical	Posterior wall, superior-medial to native orifice	Detrusor trough over intact mucosa; detrusor closed over ureter
Modified **Politano-Leadbetter** (nonrefluxing)	Transvesical	Cephalad-medial to native hiatus	Submucosal tunnel; right-angle clamp draws ureter through
Refluxing extravesical	Extravesical	Posterior dome, lateralised	Direct spatulated anastomosis
Refluxing transvesical	Transvesical	Posterior dome over a Kelly clamp	Clamp pulls ureter in

- Find the ureter at the iliac bifurcation or beneath the obliterated umbilical artery and divide it proximal to disease (no intramural dissection; frozen-section margin in stricture). Mobilise the bladder off peritoneum, dividing the ipsilateral medial umbilical ligament with or without the urachus. Test reach on the **decompressed** dome.
- **Lich-Gregoir**: a small transmural cystotomy forms the neo-orifice; incise serosa and detrusor toward the bladder neck sparing mucosa; do a spatulated 4-0 anastomosis, stent, and close detrusor over the ureter with 2-0.
- **Politano-Leadbetter**: through an anterior cystotomy, make a mucosal neo-orifice about 1.5 cm superior-medial to the native one, raise a submucosal tunnel, and draw the ureter through with a right-angle clamp; anastomose, stent, close.
- For a poorly vascularised ureter, use a non-transecting side-to-side refluxing anastomosis.
- Ureteric sutures are full-thickness; bladder bites take mucosa plus some detrusor; knots lie outside the lumen. The stent is a **6 Fr × 26–28 cm double-J**.
- Preserve periureteric adventitia (no skeletonisation); extra mobilisation buys length but devascularises.
- Primary reimplantation suits defects up to **4 cm**; longer gaps need a psoas hitch or Boari flap; renal descent adds up to **5 cm**. More bladder reach: divide the contralateral medial umbilical ligament, free peritoneum off the dome, ligate the contralateral pedicle.
- Enter **superior-medially, toward the bladder neck**; lateral sites kink on filling.
- Open is preferred for trauma and remains gold standard; laparoscopic and robotic surgery give less blood loss and shorter stay with equal patency.
- Aftercare: Foley for 1 week with a cystogram; stent for **≈6 weeks**.
- For a **leak**, drain the urinoma and keep Foley and stent; after stent removal, re-stent; if refractory, nephrostomy.
- **Late obstruction** usually follows stent removal (ischaemia, tension, narrow neo-orifice, poor spatulation, radiation, lateral site). Image antegrade and retrograde; treat short strictures endoscopically and high-grade ones by redo.
- Others: UTI, fistula, spasm, stent colic.

> **KEY POINT**
> Reimplantation fails from ischaemia or tension. Every rule (preserve adventitia, spatulate, enter superior-medial toward the bladder neck, bridge rather than stretch) prevents one or the other. The robot mainly eases intracorporeal suturing.

7 · Bridging distal ureteral defects

Why this matters: when the ureter will not reach, the bladder is brought up to it.

- P3 Boari flap — *reconstructive ladder, Boari dimensions (base ≥4 cm, apex ≥3 cm, ≤3:1, measured empty, reach up to 15 cm), psoas hitch nerves, contraindications, advancement flap*

Not in the paper answer:

- The choice of hitch, flap or both is made on the operative findings.
- Nonrefluxing Boari length is the defect plus about 3 cm (3 cm overlap, 3 cm tunnel); longer flaps need a wider base.
- A hitch cystotomy shorter than half the dome limits ascent; opening the contralateral endopelvic fascia adds mobility. The superolateral dome is chosen because it is relatively immobile.
- Ureteroureterostomy is another option beyond reach.

Option	Reach	Rule
Primary reimplant	≤4 cm	Toward bladder neck
Psoas hitch	Gains about 5 cm	Sutures superficial, parallel to the fibres; reimplant before tying
Boari flap	Up to **15 cm**	Base ≥ **4 cm**, apex ≥ **3 cm**, length:base ≤ **3:1**

- Hitch plus flap shortens the flap and preserves capacity. Hitch: oblique anterior cystotomy across about half the dome, and 4–6 2-0 sutures into the psoas minor tendon above the iliac bifurcation, avoiding the **genitofemoral** (surface) and **femoral** (deep) nerves; reimplant on the superolateral dome.
- A bladder advancement flap (transverse incision one-third of the way from dome to neck) gives reach comparable to Boari and is simpler. A **small-capacity, poorly compliant bladder** contraindicates both hitch and flap.
- Hitch mobilisation divides the contralateral vas or round ligament, with or without the contralateral superior vesical artery; refluxing reimplant if the ureter is short; 4-0 adventitial anchoring sutures.
- The Boari flap is based on the diseased side, anchored to psoas and tubularised in two layers; if still short, dissect more ureter or bring the kidney down.
- Minimally invasive: hitch to psoas minor superolateral to the iliac vessels, refluxing extravesical reimplant, flap outlined by methylene-blue distension and cut from about 2 cm above the bladder neck, 7 Fr stent.
- Complications: femoral or genitofemoral neurapraxia, iliac vessel injury, ischaemic stricture of a narrow-based flap, reduced capacity with urgency and frequency. Beyond reach: transureteroureterostomy, ileal ureter and autotransplantation.

> **KEY POINT**
> The bladder is the donor tissue: hitch first, flap second. Every flap rule (base width, 3:1, measure empty) protects the flap's blood supply and bladder capacity.

8 · Enterocystoplasty (augmentation)

Why this matters: augmentation protects the kidneys from high bladder pressure.

- Indication: neurogenic bladder with poor compliance **despite** clean intermittent catheterisation (CIC), anticholinergics, botulinum toxin and sacral neuromodulation, especially with upper tract deterioration or incontinence with skin breakdown.
- Bowel lowers pressure only if **detubularised**, which abolishes peristalsis.
- Work-up: video-urodynamics (detrusor leak point pressure predicts renal risk), serum chemistry, B12, fat-soluble vitamins (renal failure worsens acidosis), bowel history, and **confirm the patient can catheterise**. Metabolic effects: *See Ch 133*.
- Add as needed: sling, artificial urinary sphincter (AUS) or bladder neck closure, a catheterisable stoma, or a nonrefluxing reimplant for high-grade reflux.

Segment	Note
Ileum (commonest)	25 cm taken **≈20 cm** proximal to the ileocaecal valve (spares B12 and bile-salt absorption); acidosis risk
Ileocaecal	When a catheterisable channel is wanted
Sigmoid	Colonoscopy first

Segment	Note
Stomach	Possible; needs a higher incision

ALGORITHM

1. Isolate ileum and check reach to the dome.
2. Transilluminate mesentery, staple, keep a broad mesenteric base.
3. Side-to-side bowel anastomosis; close the window.
4. **Detubularise** on the antimesenteric border and fold into a U.
5. **Bivalve the bladder widely** and make a wide-mouthed anastomosis.
6. Suprapubic tube with or without Foley; leak-test; drain.

- A narrow mouth creates a poorly emptying pseudodiverticulum.
- Early: extravasation (poor suturing, flap ischaemia; preserve the arcade), UTI, bowel injury, fistula, intra-abdominal infection.
- **Perforation** follows missed catheterisation and is often masked by sensory loss. Diagnose with CT cystogram and proceed to **emergency exploration**.
- Stones are the commonest long-term complication. Neoplasia risk means surveillance endoscopy from **10 years**. Irrigation may reduce mucus; check metabolic and vitamin bloods routinely. Continence and satisfaction are high, but secondary procedures are common.
- Urethra not catheterisable: continent ileocaecocystoplasty. Patient unable to catheterise: incontinent ileovesicostomy.

KEY POINT

Augmentation exchanges a pressure problem for a lifelong bowel-in-bladder problem. Detubularisation and a wide-mouthed anastomosis win the pressure battle; catheterisation, irrigation, bloods and endoscopic surveillance are the price.

9 · Continent catheterisable channels

Why this matters: a channel lets patients who cannot catheterise the urethra empty the bladder.

- P2 Principles of continence in continent diversion; Mitrofanoff principle — *submucosal tunnel flap valve, appendix first choice, Yang-Monti alternative, channel catheter 2–4 weeks, continence >95%, stomal stenosis commonest*

Not in the paper answer:

- Minimally invasive appendicovesicostomy is widely reported, mainly in children; a minimally invasive Monti is rare and demanding.
- The caecal cuff also widens the stoma, and mobilising the caecum and ascending colon gives extra length.
- The bladder is drained (Foley or suprapubic) for 2–4 weeks before channel catheterisation starts.
- Assess dexterity, habitus, neurology, bowel history (inflammatory bowel disease, radiation, appendicectomy), renal function and video-urodynamics. The channel must reach the stoma without tension, angulation or ischaemia.
- They suit adequate-capacity bladders (neuropathic, refractory stricture) after failed urethral CIC; augment if compliance is poor. Always have a fallback if the appendix is absent or short.
- A Monti uses 2–4 cm ileum; a double Monti risks channel diverticula, a spiral Monti more revisions. Vas, colon and stomach channels do poorly.

ALGORITHM

Minimally invasive appendicovesicostomy

1. Detach the appendix with a caecal cuff and spatulate it.
2. Make a ≈4-cm submucosal detrusor tunnel; anastomose with 4-0 polydioxanone.
3. Bring the stoma to the right lower quadrant or umbilicus.
4. Drain the bladder for **2–4 weeks**. Continence is **>95%**.

KEY POINT

Use the appendix when present and long enough; reconfigure ileum when not. Continence comes from tunnel length through detrusor, and missed catheterisation endangers the bladder, not just the channel.

10 · Vesicovaginal fistula (VVF) †

Why this matters: a missed ureteric injury defeats a VVF repair.

- P3 P2 VVF: management options, early vs delayed repair — *causes, double-dye test, CT urography, catheter trial, timing, vaginal (Latzko, flap-split) vs abdominal (O'Conor) routes, interposition flaps*

Not in the paper answer:

- Outside source notes (†): leakage usually starts 1–3 weeks after hysterectomy, and a coexisting ureteric injury is present in about 10%.
- A ureterovaginal fistula is treated with a ureteric stent first; if that fails, do a ureteroneocystostomy with or without a psoas hitch.

★ Putting it together

A · NMIBC: risk and treatment

Why this matters: in non-muscle-invasive bladder cancer (NMIBC), risk group sets adjuvant therapy and surveillance. AUA/SUO risk is assigned after complete transurethral resection (TURBT).

Risk	Features	Adjuvant	Cystoscopy
Low	Low-grade (LG) solitary Ta ≤3 cm; PUNLMP	Single post-op instillation	3 mo, then yearly
Intermediate	LG Ta recurring <1 yr/>3 cm/multifocal; LG T1; high-grade (HG) Ta ≤3 cm	6-week induction mitomycin C (MMC)/gemcitabine or BCG; maintain 1 yr	3–6-monthly for 1–2 yr
High	HG T1; HG Ta >3 cm/multifocal/recurrent; CIS; variants; lymphovascular invasion (LVI); HG prostatic urethra	Re-TURBT (T1); BCG with **maintenance 3 yr** (SWOG); early radical cystectomy (RC) if highest-risk	3–4-monthly for 2 yr, then 6-monthly for 2 yr, then yearly; upper tracts yearly for 2 yr
BCG-unresponsive	HG despite adequate BCG (≥5/6 induction + ≥2/3 maintenance)	RC; if unfit: pembrolizumab (CIS), nadofaragene, gem–docetaxel	—

B · MIBC: cystectomy vs preservation *See Ch 131, 132*

Why this matters: in muscle-invasive disease (MIBC), selected patients can keep the bladder.

	NAC + RC + pelvic node dissection	Trimodality therapy
Who	Default if fit; neoadjuvant chemotherapy (NAC: gem–cis or ddMVAC) if cisplatin-eligible	Solitary T2, complete TURBT, **no CIS/hydronephrosis**, good function; or unfit
Cisplatin-ineligible	GFR <60, ECOG ≥2, grade ≥2 hearing loss or neuropathy, or NYHA III–IV: local therapy	
After	Adjuvant nivolumab (CheckMate 274): ≥pT3/N+ after NAC, or no NAC	Lifelong cystoscopy; salvage RC 10–30%

C · Choosing the diversion *See Ch 133–135*

Why this matters: renal function, urethra and catheterising ability decide the diversion.

	Ileal conduit	Orthotopic neobladder	Continent cutaneous
Best for	Frail, poor renal function, cannot catheterise, neurological	Default continent option, either sex	Unusable urethra (margin tumour, stricture) or sphincter
Must have	Usable (non-irradiated) ileum	eGFR **>35–40**; intact sphincter; negative urethral frozen section; will catheterise	eGFR >35–40; dexterity; efferent limb (appendix, tapered ileum)
Main problems	Stoma problems, ureteroileal stricture	Night leakage; late retention (women); acidosis	Stomal stenosis, catheterisation difficulty, stones

D · Metabolic complications by segment

Why this matters: each segment gives a typical electrolyte pattern.

Segment	Disturbance	Treatment
Ileum / colon	**Hyperchloraemic metabolic acidosis** (ammonium and chloride absorbed)	Bicarbonate, potassium citrate; chlorpromazine/nicotinic acid
Jejunum	**Hyponatraemic hypochloraemic hyperkalaemic acidosis**	Sodium chloride, thiazide; avoid jejunum
Stomach	**Hypochloraemic hypokalaemic alkalosis**; haematuria–dysuria syndrome	H2 blocker/PPI

Losing the terminal ileum lowers B12 and causes bile-salt diarrhoea. Acidosis is buffered by bone, causing osteomalacia. *See Ch 133*

E · Metastatic urothelial carcinoma

Why this matters: fitness for platinum decides first-line therapy.

ALGORITHM

1. **Fitness**: cisplatin-, platinum-only or platinum-ineligible.
2. **First line**: enfortumab vedotin (EV) with pembrolizumab (EV-302); otherwise gem–cis (gem–carbo if cisplatin-ineligible).
3. **If no progression on platinum**, give avelumab maintenance (JAVELIN Bladder 100).
4. **Later lines**: pembrolizumab; erdafitinib (if FGFR-altered); EV; sacituzumab govitecan. If platinum-ineligible, pembrolizumab first. *See Ch 131*